Fiscal-Mobility Propagation in Currency Unions*

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30th October 2026
[Latest version]

Abstract

This paper argues that labour mobility shapes shock propagation in currency unions through its interaction with regional fiscal space. In a New Keynesian open-economy model of migration with debt limits, I show analytically that mobility has two opposing effects on fiscal space: emigration expands fiscal capacity through spillovers that raise per-capita fiscal contributions, but contracts it through increased per-capita debt burden. The dominant effect on constrained governments determines whether mobility enables a more stabilising fiscal response to negative idiosyncratic shocks—or deepens pro-cyclical consolidations that amplify the original shock. Using LinkedIn data, I construct a novel panel of monthly intra-EU migration flows for 2000–2019. Estimated SVAR migration shocks applied to local projections show that a a one-standard-deviation emigration shock contracts government consumption during high-debt regimes by a cumulative 1.7 percentage points of trend GDP over five years relative to low-debt regimes, consistent with the amplification mechanism dominating where the debt burden channel is strongest.

Keywords: Labour mobility; fiscal space; fiscal policy; currency unions

JEL Classification: E24, E32, E62, F22, F45

*For valuable advice and supervision, I thank Ricardo Reis. I also thank Sasha Abrahams, Boris Beyen, Matthias Doepke, Marina Feliciano, Jonathon Hazell, Ethan Ilzetzki, Corentin Laffitte, Alan Manning, Benjamin Moll, Rachel Ngai, Tommaso Paolucci, Alessandro Sciacchetano, Liliana Varela, and seminar participants at LSE for valuable comments, discussions, and suggestions. The project that gave rise to these results received the support of a fellowship from "la Caixa" Foundation (ID 100010434), under the fellowship code LCF/BQ/EU24/12060047.

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I Introduction

Does labour mobility always stabilise regional imbalances in a currency union? The conventional wisdom, going back to the seminal work of Mundell (1961), would respond with an emphatic *yes*—when monetary policy is shared and the possibility of internal exchange rate adjustments is ruled out, labour mobility takes up the stabilising role of attenuating regional imbalances. The subsequent optimal currency area (OCA) literature identified fiscal policy as a second key stabilisation instrument substituting for the lost monetary independence (Kenen, 1969; Galí and Monacelli, 2008). The consensus view is, therefore, that labour mobility and fiscal policy are two separate stabilisation mechanisms aiding members of a currency union facing idiosyncratic shocks.

In this paper, I argue that the fiscal and the mobility dimensions are not independent of each other, and in particular that they interact via the fiscal space of regional governments. The result of this interaction affects how idiosyncratic shocks propagate in the currency union, a mechanism I call *fiscal-mobility propagation*. Depending on how migration affects fiscal space, fiscal-mobility propagation can reinforce the stabilising role of labour mobility or amplify the original regional imbalances, making the role of labour mobility in a currency union inseparable from the fiscal conditions of its members.

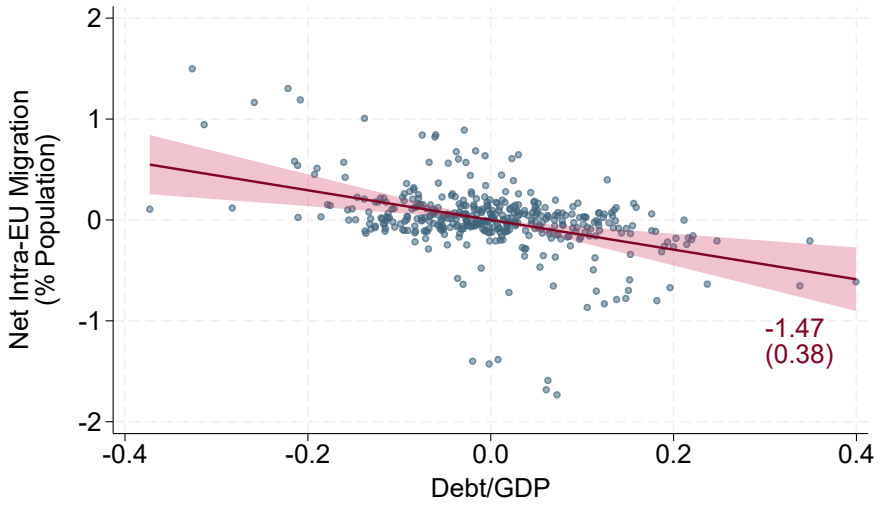
Fiscal space, defining the capacity for governments to pursue their desired fiscal policy, is the key link connecting labour mobility and fiscal policy. A fiscally constrained government may be forced into consolidations that depress economic activity and wages, raise taxes and deteriorate public services, incentivising the labour force to emigrate. However, depending on how emigration itself feeds back into fiscal space, the extent of such consolidations can be milder or harsher than in the absence of migration, affecting the severity of the recession. Intuitively, the impact of emigration on the government's fiscal capacity can go both ways. Emigration may tighten the constraints faced by the government by eroding the tax base and spreading the burden of government expenditures, as well as that of the existing public debt, over fewer taxpayers. That said, emigration can also provide a welcome fiscal relief if its composition reduces unemployment benefit transfers, or if the reduction in labour supply spills over into higher wages for those who remain, raising per-capita net fiscal contributions in the origin region. The net effect of migration on fiscal space is therefore not obvious, but its importance is central to understanding the role of labour mobility in currency unions, as it determines whether fiscal policy is allowed to be more expansive than otherwise, or whether contractionary consolidations must be reinforced.

This paper builds a tractable New Keynesian open-economy model to formalise the opposing forces that migration exerts on fiscal space, characterises the resulting stabilising or amplifying fiscal-mobility propagation mechanisms, and uses novel monthly data on migration to assess which one holds empirically.

The motivation for this paper is the 21st-century European Union (EU), and the sovereign debt crisis of the 2010s in the Euro Area. The introduction of the euro in 1999 established a

currency union among a continuously expanding number of European countries, but its setup comprises institutional features that distinguish it from other currency unions such as the US, and make the relevance of fiscal-mobility interactions all the more salient. First, despite its single market, business cycles in the EU are not fully synchronised (Frankel and Rose, 1998), giving rise to regional imbalances. Second, labour mobility alone is insufficient to stabilise these imbalances (Beyer and Smets, 2015)—for comparison, less than 5% of working-age EU citizens reside in a different member state than their country of birth, compared to around 30% of Americans living outside their state of birth (Arpaia et al., 2016)—creating the need for fiscal stabilisation. Third, despite early warnings of the need for fiscal integration alongside the monetary union (the MacDougall Report, 1977; the Delors Report, 1989), a broad federal budget for the EU was never set up, leaving the required fiscal response of individual member states vulnerable to contractions in their respective fiscal space.

FIGURE 1: Migration and Public Debt in the EU (2000–2019)



Source: Eurostat, author's calculations.
Notes: Each dot corresponds to a country-year pair. Sample includes all EU-28 countries (including the United Kingdom) in 2000–2019, for all available data. Both variables are residualised from country and year fixed effects. Solid red line shows fitted regression, and shaded area shows 95% confidence band. The regression slope coefficient and its standard error clustered at the country level (in parentheses) are displayed.

These institutional fragilities were exposed during the sovereign debt crisis, in which high levels of public debt were coupled with abnormally large levels of intra-EU migration. As evidenced by Figure 1, there is a strong correlation between labour mobility and fiscal space for EU members during this period. The crisis in Euro Area peripheral economies led debt levels to breach the Maastricht limit of 60% of GDP, and tested the limits of market borrowing, rendering market-implied debt constraints binding for these countries. Some of them (Greece, Portugal, Ireland, Spain, Cyprus) received Troika bailouts or other forms of assistance that conditioned the respective governments to enact austerity programmes. The resulting fiscal consolidations had the effect of further contracting economic activity which was already severely depressed. At the same time, these highly indebted economies saw an exodus of working-age population to northern European countries—in Greece alone, annual emigration flows to other European countries more than doubled during the crisis,

from 0.26% of its population in 2009 to 0.70% in 2012. Policy makers at the time viewed the austerity programmes as a necessary evil, and the mass emigration as a separate natural economic outcome which aided the stabilisation procedure.

An overlooked question is how mass emigration itself impacted the required extent of the fiscal consolidation programmes. Namely, if emigration from southern to northern EU countries provided a fiscal relief to indebted countries, then labour mobility may have alleviated the required severity of the pro-cyclical fiscal consolidations, and fiscal-mobility propagation had a stabilising effect. By contrast, if emigration from South to North eroded the tax base and shrank fiscal space even further, then southern European governments may have been forced to consolidate their fiscal positions by more than in the counterfactual scenario of no migration, and in doing so triggered the fiscal-mobility amplification mechanism that deepened the recession beyond what initial austerity alone would have implied. More broadly, this paper asks whether labour mobility stabilises or amplifies idiosyncratic shocks through its interaction with fiscal space, and which of the two fiscal-mobility propagation mechanisms characterises the European experience during the sovereign debt crisis.

I answer these research questions by first developing a New Keynesian open-economy model of migration. The model is an adaptation of the static external demand imbalances model of labour mobility introduced in Farhi and Werning (2014), extended to an infinite horizon to allow for government debt accumulation. The model features two initially symmetric regions in a currency union with independent fiscal authorities. The regions are hit by asymmetric demand shocks, and workers have an incentive to migrate to the region with the higher real wage, subject to a congestion cost and location preference shocks. As in Farhi and Werning (2014), this migration generates a positive spillover effect on the origin region—since the existence of trade implies that regional aggregate demand falls by less than population—which helps attenuate regional imbalances as envisioned by Mundell (1961). Regional governments can also use fiscal policy to further attenuate the effects of their idiosyncratic shocks, but they are subject to an exogenous stochastic per-capita real debt limit. Nominal price rigidities imply that fiscal expansions increase real wages, incentivising immigration, while consolidations depress them, and incentivise emigration.

The debt limit imposes an upper bound on how much regional governments can spend per capita in response to idiosyncratic shocks, defining the government's fiscal space. In the model, emigration exerts two opposing forces on fiscal space. On the one hand, the migration spillover effect will also increase the wages of the workers who remain. Even without modelling unemployment, emigration therefore increases per-capita fiscal contributions. On the other hand, emigration erodes fiscal capacity by spreading any expenses that do not scale with population over fewer workers. Even if government spending scales one-for-one with population—that is, if the government chooses a *per-capita* rather than an *aggregate* level of spending—legacy debt does not scale, so emigration necessarily increases the per-capita debt burden. This debt burden channel is analogous to the $r - g$ condition of the public finance literature (Domar, 1944; Blanchard et al., 1991; Blanchard, 2019; Mehrotra and Sergeyev, 2021),

with population (de)growth taking the place of GDP growth.

Under additional simplifying assumptions, I solve the model fully in closed form, and I show that the key coefficients keep their signs when these assumptions are relaxed, so the results extend qualitatively. In particular, I show analytically that the model has three possible regimes, and that which of the two effects of migration on fiscal space dominates determines whether fiscal-mobility propagation is stabilising or amplifying. If the positive fiscal contribution channel dominates, emigration expands fiscal space and allows the government to raise per-capita spending to cushion the regional bust. I call this *fiscal-mobility stabilisation*, and it complements the Mundellian view of labour mobility as a stabilising force. If instead the negative debt burden channel dominates, emigration tightens fiscal constraints even more and forces the government to cut spending, which depresses regional demand further and amplifies the original shock. I call this *fiscal-mobility amplification*, and it can take two forms. Under *partial* fiscal-mobility amplification, the contraction of fiscal space amplifies shocks by less than Mundellian stabilisation dampens them. Under *full* fiscal-mobility amplification, it amplifies them by more, so that labour mobility is no longer a net stabilising force in the union. I then characterise the conditions under which each of the three regimes occurs, in terms of the relative strength of the debt burden and the migration spillover.

I test these predictions on a monthly panel of EU countries between January 2000 and December 2019. I use LinkedIn data from Revelio Labs to construct a novel monthly series of intra-EU migration, based on users' self-reported cross-border movements for new jobs and education. This monthly panel sidesteps data limitations of existing official migration statistics for the EU, which are poorly reported within the Schengen Area, yet validates well against available official bilateral flows with a strong log-log correlation of 0.83. The monthly frequency is also used for identification of a structural vector autoregression (SVAR) model using short-run timing restrictions on migration. The SVAR shows that a 1pp migration rate shock has an effect on receiving countries of reducing unemployment by roughly 0.2pp at peak and debt by roughly 1.3pp of trend GDP, and by around 0.5pp and 1pp respectively in origin countries. However, and most strikingly, falling debt in origin countries is not indicative of fiscal-mobility stabilisation, but rather the result of forced fiscal consolidations—using local projections to estimate the response of quarterly fiscal variables, I show that during high-debt periods emigration forces fiscal consolidations rather than enabling expansions. Government consumption in highly indebted countries (above 90% of trend GDP) contracts by a cumulative 1.7pp of trend GDP over 20 quarters relative to low-debt countries, in response to a 1 standard deviation emigration shock. This result is consistent with the prediction that, for a given migration spillover, higher legacy debt makes fiscal-mobility amplification more likely.

Related literature. The paper relates to several strands of the literature. First, it adds to the OCA literature, which has repeatedly emphasised the essential adjustment role of labour mobility (Mundell, 1961; Blanchard and Katz, 1992; Farhi and Werning, 2014), and my model builds upon Farhi and Werning's framework. This literature has also studied the role of

integrated and independent fiscal policies as a replacement for monetary policy in currency unions (Kenen, 1969; Galí and Monacelli, 2008; Farhi and Werning, 2017), and more recently, and in close spirit to my framing, Bellifemine et al. (2025) study how heterogeneous fiscal space affects shock propagation across members of a currency union, albeit ignoring shock asymmetry or labour mobility. Wildasin (2000) and Baglioni et al. (2016) study how labour mobility constrains fiscal adjustments in currency unions, but they do so through the lens of tax competition as opposed to fiscal space. To the best of my knowledge, this paper is the first to formalise the interaction between labour mobility and fiscal space as a determinant of shock propagation in currency unions, and to test it empirically.

A separate literature studies fiscal-mobility interactions outside OCA, but these tend to ignore the feedback between migration and fiscal space in equilibrium. Applied micro papers study the migration elasticity to tax changes (see Kleven et al. 2020 for a review). Some macro papers study the short-run interaction between fiscal consolidations and migration (Bandeira et al., 2022), but in small open economy settings that abstract from union-wide general equilibrium effects. In the opposite direction, a vast literature focuses on how migration affects public finances, but this usually takes the form of a demographic analysis of the long-run fiscal sustainability of social programmes (Borjas, 1994; Storesletten, 2000; Dustmann and Frattini, 2014; Fiorio et al., 2024; Bernardino et al., 2025). This paper fills this gap by studying how migration affects short-run fiscal capacity and the conduct of fiscal policy for business cycle management in a currency union.

The paper also relates to the literature on fiscal space and sovereign debt constraints. The notion that governments face binding debt limits, and that these limits have real macroeconomic consequences, is well established (Eaton and Gersovitz, 1981; Reinhart and Rogoff, 2010; Lorenzoni and Werning, 2019). Blanchard et al. (1991) show that the sustainability of public debt depends critically on the relationship between interest rates and growth, a condition mirrored in this paper by the analogous role of interest rates relative to population growth in determining the debt burden channel. Romer and Romer (2019) provide empirical evidence that fiscal space shapes the severity of post-crisis recessions, consistent with the amplification mechanism documented here. This paper contributes to this literature by showing that labour mobility is itself a determinant of fiscal space, creating a feedback between migration and the government's capacity to conduct stabilising fiscal policy.

Finally, following the austerity programmes applied during the European sovereign debt crisis, a long debate ensued as to the costs of austerity. The literature split into factions, where one camp argued that expenditure-based consolidations are least harmful and can even be expansionary (Alesina and Ardagna, 2010; Alesina et al., 2019), while another deemed them too costly (Blanchard and Leigh, 2013; Guajardo et al., 2014; Jordà and Taylor, 2016) with potentially persistent hysteresis effects (DeLong and Summers, 2012). More broadly, the size of fiscal multipliers has been shown to be state-dependent (Corsetti et al., 2013; Ilzetzki et al., 2013), a finding this paper extends by adding labour mobility as an additional source of state-dependence. Emigration induced by fiscal consolidations feeds back into fiscal space,

and I show that the sign of the interaction affects the overall costs of austerity.

Roadmap. The remainder of the paper proceeds as follows. Section II presents the dynamic New Keynesian model of migration and fiscal constraints. Section III provides analytical results showing the fiscal-mobility interactions and how they can be stabilising or amplifying. In Section IV I describe the data collection process, while in Section V I present my empirical strategy and the empirical results. Section VI concludes.

II Dynamic Model of Migration and Fiscal Constraints

The model presented in this section takes the canonical static labour mobility model of Farhi and Werning (2014), and extends it into an infinite horizon to allow for dynamic fiscal rules and debt accumulation. In doing so, the relevant endogenous state variable for agents becomes their history of regions of residence.

There are two regions $i \in \mathcal{I} = \{\text{North } (N), \text{South } (S)\}$ that form a currency union. Agents are allowed to freely move across the two regions, which also share a monetary authority, but each region sets its own fiscal policy. Agents make decisions over time $t > 0$, which is discrete and of infinite horizon, taking their starting position at $t = 0$ as given. At $t = 0$, the union's economy is at a fully symmetric steady-state equilibrium: regions are initially of equal size, and all prices across regions and firms are identical and normalised to 1. Shocks in the economy are unanticipated MIT shocks, meaning agents and firms expect the economy to remain at its current steady state, but realised shocks change the equilibrium path. Following each shock, agents and firms have perfect foresight over the resulting deterministic transition path.

Households. There is a constant unit mass of worker-agents split across the two regions. Let $s_t \in \mathcal{I}$ denote the region in which an agent resides at time t, and $s^t \in \mathcal{I}^{t+1}$ the history of regions of residence for the agent up to period t, including the region of origin s_0. Agents are ex-ante identical except for their origin region s_0, and all ex-post differences result from having different residence histories s^t. A worker with residence history s^t gains log-utility in each period t from per-capita consumption $c_t(s^t)$ and disutility from supplying per-capita labour $\ell_t(s^t)$.[1] Preferences are as in Greenwood, Hercowitz and Huffman (GHH), implying no wealth effects on labour supply. Formally, utility from having residence history s^t is

$$U(s^t) = \max_{\{c_t(s^t), \ell_t(s^t), a_t(s^t)\}} u_t(s^t) + \beta \, \mathbb{E}_t U(s^{t+1}), \tag{1}$$

$$u_t(s^t) = \log\left(c_t(s^t) - \psi \frac{\ell_t(s^t)^{1+\varphi}}{1+\varphi}\right) - \theta \log n_{s_t,t} + \log \xi_{s_t,t}, \tag{2}$$

[1] Unless stated otherwise, throughout the model lower-case variables are per-capita outcomes, whereas upper-case indicates aggregate variables.

with region of origin s_0 given. For notational brevity, aggregate state variables are suppressed. Despite there being institutional freedom of movement for labour, $\theta \log n_{s_t,t}$ captures congestion costs caused by excessive population $n_{s_t,t}$ in the current region. These can be thought of as limited supply in housing and other resources that get congested if regional population becomes too high, imposing a utility cost on all local residents. Furthermore, $\log \xi_{s_t,t}$ is a region-specific shock to utility, capturing agents' changes to location preferences exogenous to economic activity. Moreover, φ is the inverse Frisch elasticity, ψ is a labour supply disutility shifter, and $\theta > 0$ is the congestion cost sensitivity parameter, which affects how responsive migration is to shocks (if $\theta \to \infty$, regional population always stays constant).

There is an incomplete set of asset markets, and households can only save through purchasing risk-free government bonds $a_t(s^t)$ which are traded union-wide with nominal net return r_t. Households start with the same assets regardless of region of origin, $a_0(s^0) = a_0 \geq 0$, and their period budget constraint is

$$P_t(s_t)\, c_t(s^t) + a_t(s^t) = W_{s_t,t}\, \ell_t(s^t) + \pi_t(s^t) + (1 + r_{t-1}) a_{t-1}(s^{t-1}), \tag{3}$$

where $W_{s_t,t}$ is the wage rate in the agent's current region of residence s_t, and $\pi_t(s^t)$ are per-capita nominal profits distributed to workers with residence history s^t. $P_t(s_t)$ is the consumer price index (CPI) for an agent residing in region s_t. The consumption term $c_t(s^t)$ is an aggregate which comprises consumption of the final good from each region i, $c_t^{(i)}(s^t)$. More concretely, $c_t(s^t)$ is a constant elasticity of substitution (CES) aggregator of the form

$$c_t(s^t) = \left(\alpha_{s_t,t}^{\frac{1}{\sigma}} c_t^{(s_t)}(s^t)^{\frac{\sigma-1}{\sigma}} \; + \; (1 - \alpha_{s_t,t})^{\frac{1}{\sigma}} c_t^{(-s_t)}(s^t)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{4}$$

where the notation $-s_t$ (or $-i$) is used, throughout the model, to indicate the other region from the one being indexed. The exogenous variable $\alpha_{i,t} \in (0,1)$ captures the degree of home bias in consumption. Crucially, it is allowed to vary across regions and time, and this is the source of demand imbalance shocks to the model. Under this aggregation, the consumer price index is in turn given by

$$P_t(s_t) = \left(\alpha_{s_t,t} P_{s_t,t}^{1-\sigma} + (1 - \alpha_{s_t,t}) P_{-s_t,t}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \tag{5}$$

where $P_{i,t}$ is the price of the final good produced in region i, to be defined shortly. A no-Ponzi condition applies.

Agents' first-order conditions pin down per-capita labour supply

$$\ell_t(s^t) = \ell_t(s_t) = \left(\frac{W_{s_t,t}}{\psi P_t(s_t)} \right)^{\frac{1}{\varphi}}, \tag{6}$$

the Euler equation

$$1 = \beta \mathbb{E}_t (1 + r_t) \frac{P_t(s_t)}{P_{t+1}(s_{t+1})} \frac{\tilde{c}_t(s^t)}{\tilde{c}_{t+1}(s^{t+1})}, \tag{7}$$

where $\tilde{c}_t(s^t) \equiv c_t(s^t) - \psi \frac{\ell_t(s^t)^{1+\varphi}}{1+\varphi}$, and per-capita consumption of each region's final good,

$$c_t^{(s_t)}(s^t) = \alpha_{s_t,t} \left(\frac{P_{s_t,t}}{P_t(s_t)} \right)^{-\sigma} c_t(s^t) \quad \text{and} \quad c_t^{(-s_t)}(s^t) = (1 - \alpha_{s_t,t}) \left(\frac{P_{-s_t,t}}{P_t(s_t)} \right)^{-\sigma} c_t(s^t). \tag{8}$$

Notice that, due to GHH preferences, there are no wealth effects on labour supply, which depends only on the current wage rate and CPI, both of which are independent of prior residence history. Therefore, labour supply does not depend on the entire history of regions of residence s^t, but on the current place of residence s_t alone, which simplifies the problem.

Migration and population. Agents freely choose their region of residence s_t for each period, taking as given their prior history of residence, in order to maximise utility. More concretely, they choose

$$s_t = \underset{\widehat{s}_t \in \mathcal{I}}{\operatorname{argmax}}\, U(\{s^{t-1}, \widehat{s}_t\}), \quad s^{t-1} \text{ given}, \tag{9}$$

where agents also take as given the overall population distribution of workers across the two regions, affecting aggregate variables and congestion costs. The probability mass function of this distribution is the mass of agents who have residence history s^t at any point in time and is denoted by $n(s^t)$. Union-wide population is fixed and the population constraint satisfies

$$\sum_{s^t \in \mathcal{I}^{t+1}} n(s^t) = 1, \quad \forall t. \tag{10}$$

For notational brevity, the population mass in region i at time t is denoted as $n_{i,t} \equiv \sum_{s^t | s_t = i} n(s^t)$. The symmetric starting point implies $n_{N,0} = n_{S,0} = \frac{1}{2}$.

Firms. Each region has a competitive final good producer that aggregates a continuum of non-tradable intermediate varieties indexed by $j \in [0,1]$ with a CES $\nu > 1$ production technology

$$Y_{i,t} = \left(\int_0^1 Y_{ij,t}^{\frac{\nu-1}{\nu}} \mathrm{d}j \right)^{\frac{\nu}{\nu-1}}. \tag{11}$$

The final good firm minimises its expenditure subject to the above production technology, resulting in a demand for each variety given by

$$Y_{ij,t} = \left(\frac{P_{ij,t}}{P_{i,t}} \right)^{-\nu} Y_{i,t}, \tag{12}$$

and a competitive price of final goods

$$P_{i,t} = \left(\int_0^1 P_{ij,t}^{1-\nu} \mathrm{d}j \right)^{\frac{1}{1-\nu}}. \tag{13}$$

Each variety is in turn produced by an intermediate monopolist that hires labour $L_{ij,t}$ in competitive labour markets at a nominal wage rate $W_{i,t}$, and intermediate firms need to pay a payroll tax rate $\tau_{i,t}$ on top of this. Labour is used to produce intermediate varieties in a constant returns to scale production function

$$Y_{ij,t} = A\, L_{ij,t}, \tag{14}$$

with A representing aggregate productivity, which is symmetric across regions and constant. Intermediate firms set prices monopolistically, but they do so one period ahead, meaning their period t price is set at time $t-1$ before the realisation of any period t shocks. In period t, firms can only adjust next period's price, following

$$P_{ij,t+1} = \frac{\nu}{\nu - 1} \mathbb{E}_t \frac{(1+\tau_{i,t+1}) W_{i,t+1} L_{ij,t+1}}{Y_{ij,t+1}}. \tag{15}$$

This setup implies that prices are flexible dynamically and to anticipated shocks, but on impact they suffer from complete within-period stickiness relative to any unanticipated shock. The main advantage of this form of price setting is that it provides a very clean form of aggregate labour demand, useful for analytical purposes, but the model could easily instead incorporate more standard forms of nominal price rigidity—such as Rotemberg (1982) or Calvo (1983) pricing—at the expense of analytical tractability. Namely, due to symmetry, all intermediate firms are identical and employ the same amount of labour, implying that aggregate production in the region is simply equal to

$$Y_{i,t} = A\, L_{i,t}, \tag{16}$$

where $L_{i,t} \equiv \int_0^1 L_{ij,t} \mathrm{d}j$. Since prices are within-period sticky, all firms demand whatever amount of labour they need to satisfy product demand at the pre-determined prices, conditional on making non-negative profits. I impose that productivity A is high enough so that this is always the case. Aggregate labour demand then takes the clean form of

$$L_{i,t} = \frac{Y_{i,t}}{A}. \tag{17}$$

Aggregate nominal profits in each region are in turn given by

$$\Pi_{i,t} = P_{i,t} Y_{i,t} - (1+\tau_{i,t}) W_{i,t} L_{i,t}. \tag{18}$$

For simplicity, I assume that firms are equally owned by all households in the union, regardless of where they reside. This can be institutionalised through a union that allows for cross-border firm ownership. Then, per-capita profits are equally distributed to all workers in the union and given by

$$\pi_t(s^t) = \pi_t = \Pi_{N,t} + \Pi_{S,t}. \tag{19}$$

This results in all agents receiving the same dividend income regardless of their region of residence, and labour income becomes the only source of income differentials across regions. This setup is particularly useful as it implies that regional real wage differentials are the only income incentive for migration.

Regional governments. There is no fiscal union alongside the currency union, and each region i has its own regional government. The regional government uses its labour income tax to finance a per-capita level of consumption of final goods $g_{i,t}$. Government spending is "thrown into the ocean", providing no direct utility to households. Furthermore, for simplicity, I assume that the government is entirely home-biased, implying that it only buys final goods from its own region.

The regional government targets a counter-cyclical fiscal rule, raising spending and cutting taxes during busts, and reversing this during booms, according to

$$g_{i,t} = g_0 - \rho_g(y_{i,t} - y_0) + \zeta^g_{i,t} \qquad \text{and} \qquad \tau_{i,t} = \tau_0 + \rho_\tau(y_{i,t} - y_0) + \zeta^\tau_{i,t},$$

where $\rho_g, \rho_\tau > 0$ govern the response of spending and taxes, $y_{i,t}$ is per-capita real output, and g_0, τ_0, y_0 denote the initial symmetric steady-state values of per-capita real government spending, labour taxes, and per-capita real output, respectively. The desired level of spending and taxation is, furthermore, subject to policy shocks $\zeta^g_{i,t}$ and $\zeta^\tau_{i,t}$, which may originate for instance from exogenous political economy reasons.

Debt constraints, introduced shortly, may prevent the government from following this rule exactly in any given period, or doing so today may compromise following it in the future. The government therefore treats the rule as a benchmark, choosing a path of per-capita real government spending $\{g_{i,t}\}_{t=1}^{\infty}$, labour taxes $\{\tau_{i,t}\}_{t=1}^{\infty}$, and per-capita end-of-period nominal debt $\{b_{i,t}\}_{t=1}^{\infty}$ that minimises quadratic deviations from its target,

$$\begin{aligned}\min_{\{g_{i,t}\geq 0,\, \tau_{i,t},\, b_{i,t}\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} \Big\{ &\big[g_{i,t} - \big(g_0 - \rho_g(y_{i,t} - y_0) + \zeta^g_{i,t}\big)\big]^2 \\ &+ \lambda \big[\tau_{i,t} - \big(\tau_0 + \rho_\tau(y_{i,t} - y_0) + \zeta^\tau_{i,t}\big)\big]^2 \Big\},\end{aligned} \tag{20}$$

with λ being the relative weight the government puts on the tax rule. In choosing its fiscal variables, the government is subject to a period budget constraint which can be written in per-capita nominal terms as

$$P_{i,t} g_{i,t} + (1 + r_{t-1}) \frac{n_{i,t-1}}{n_{i,t}} b_{i,t-1} = \tau_{i,t} W_{i,t} \ell_{i,t} + b_{i,t}, \tag{21}$$

in which $\ell_{i,t}$ is per-capita labour supply from the household optimisation problem (which is equal for all residents of region i at time t regardless of residence history, as noted before) and starting per-capita debt stock is given and equal for both regional governments, $b_{i,0} = b_0 \geq 0$. A no-Ponzi condition applies, and the government is further constrained by an exogenous

per-capita real debt limit $\bar{b}_{i,t} \geq b_0$, which is stochastic and realised at the start of period t before the government chooses its fiscal variables. Because debt is denominated in nominal terms, the limit requires that the real value of debt, deflated by the price of the region's final good, does not exceed $\bar{b}_{i,t}$:

$$\frac{b_{i,t}}{P_{i,t}} \leq \bar{b}_{i,t}. \tag{22}$$

Endogenising or microfounding this debt limit is beyond the scope of this paper, but it can be interpreted as either being imposed union-wide institutionally (such as deficit and debt limits enshrined in the Maastricht Treaty), or imposed by some exogenous market force (see Eaton and Gersovitz 1981; Arellano 2008; Ghosh et al. 2013 for ways to endogenise $\bar{b}_{i,t}$). Shocks to $\bar{b}_{i,t}$ can then be interpreted as either exogenous changes to the union's institutional fiscal limits, or exogenous changes to market sentiment affecting government access to bond markets and limits on borrowing.

Monetary authority. The currency union has a common monetary authority that sets the path of nominal interest rates r_t. This central bank follows a Taylor rule that targets deviations in union-wide CPI inflation and output from their initial symmetric steady-state values,

$$r_t = \frac{1}{\beta} + \phi_P \frac{P_t - P_{t-1}}{P_{t-1}} + \phi_Y \frac{Y_t - Y_0}{Y_0} - 1, \tag{23}$$

where union-wide CPI and output are population-weighted averages of their regional counterparts, given by

$$P_t = \left(n_{N,t} P_t(N)^{1-\sigma} + n_{S,t} P_t(S)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad \text{and} \quad Y_t = \left(n_{N,t}^{\frac{1}{\sigma}} Y_{N,t}^{\frac{\sigma-1}{\sigma}} + n_{S,t}^{\frac{1}{\sigma}} Y_{S,t}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \tag{24}$$

with $\phi_P > 1$ satisfying the Taylor principle and $\phi_Y > 0$.

Equilibrium. Given initial prices and allocations in the symmetric steady-state equilibrium

$$\{P_0(s_0) = P_{i,0} = P_{ij,0} = P_0 = 1, a_0(s^0) = b_{i,0} = a_0 = b_0 \geq 0, g_{i,0} = g_0, \tau_{i,0} = \tau_0, n_{i,0} = n_0 = \tfrac{1}{2}\},$$

initial regions of residence s_0, and exogenous random variables $\{\xi_{i,t}, \alpha_{i,t}, \zeta^g_{i,t}, \zeta^\tau_{i,t}, \bar{b}_{i,t}\}$, an equilibrium for $t > 0$ is a set of per-capita allocations $\{c_t(s^t), c_t^{(i)}(s^t), \ell_t(s_t), a_t(s^t), \pi_t(s^t), g_{i,t}, b_{i,t}\}$, aggregate allocations $\{L_{i,t}, L_{ij,t}, \Pi_{i,t}, Y_{i,t}, Y_{ij,t}, Y_t\}$, a system of prices and taxes $\{W_{i,t}, P_t(s_t), P_{i,t}, P_{ij,t}, P_t, \tau_{i,t}, r_t\}$, residence histories s^t, and population masses $n(s^t)$, such that conditions (1)–(24) are satisfied, Ponzi schemes are ruled out, and markets clear in every period and region.

The goods market clearing condition in region i is

$$Y_{i,t} = \sum_{s^t \in \mathcal{I}^{t+1}} n(s^t) c_t^{(i)}(s^t) + n_{i,t} g_{i,t}. \tag{25}$$

Labour markets clear whenever aggregate labour demand equals aggregate labour supply in each region i, reading

$$L_{i,t} = n_{i,t}\ell_t(i). \tag{26}$$

Union-wide assets market clearing requires

$$n_{N,t}b_{N,t} + n_{S,t}b_{S,t} = \sum_{s^t \in \mathcal{I}^{t+1}} n(s^t)a_t(s^t). \tag{27}$$

III Fiscal-Mobility Propagation

III.A Additional assumptions and auxiliary results

This section shows the key theoretical results of the paper, regarding how fiscal policy and labour mobility interact with each other and propagate shocks. To show these results, I make additional simplifying assumptions to the model that allow for greater analytical tractability. To show the results hold more generally, the model can be solved using numerical methods.

First, this section only analyses the propagation of shocks throughout the economy on impact, and at a symmetric equilibrium. Given that shocks are completely unanticipated by firms (MIT shocks) and that prices are pre-determined due to the one-period lag in their setting, on impact this implies prices are fully sticky at their initial symmetric steady-state level, normalised to 1, and can hence be suppressed throughout this section. Under more conventional Calvo or Rotemberg pricing, partial price stickiness applies and the results derived in this section also apply along the transition path. The choice of period-ahead price setting is, as noted before, purely motivated by analytical tractability.

Second, with no price changes, I assume that on impact the central bank just focuses on stabilising union-wide output. More concretely, the central bank sets r_t such that union-wide aggregate demand Y_t remains completely fixed in response to any shock. Besides helping pin down consumption responses, this property is also useful as it implies that any effects of shocks or fiscal policy simply redirect economic activity and population from one region to another, allowing us to abstract from global size effects.

Third, to simplify the government's problem, I assume that, due to political economy constraints, governments cannot adjust their tax rates $\tau_{i,t}$, which remain fixed at their initial symmetric steady-state τ_0 level, and fiscal policy is enacted through government spending $g_{i,t}$ alone. This assumption ends up not affecting the impact analysis of this section, as at a symmetric equilibrium migration would not respond even if taxes were allowed to change. Due to pre-determined prices, labour demand is perfectly inelastic on impact and the economic incidence of the labour tax is fully borne by firms, leaving labour supply decisions unchanged. The labour tax is, therefore, fully absorbed into lower profits, which are equally distributed as lump-sum to all households regardless of residence. At an equilibrium with symmetric wealth distributions across regions, this causes an equal wealth effect in both regions, and relative aggregate demands remain unchanged. The incentive to migrate is, for

this reason, unaffected by taxes on impact. Formalising,

$$\frac{\mathrm{d}n_{i,t}}{\mathrm{d}\tau_{i,t}} = 0. \tag{28}$$

A complete proof of this result is provided in Appendix A.1.

PROPOSITION 1 (Migration spillovers and the Mundellian channel)**.** *Keeping fiscal policy fixed and symmetric across regions, emigration strictly increases per-capita output in the origin region, to first order at a symmetric equilibrium. The result requires either imposing Keynesian-cross stability or sufficient parametric restrictions on home bias.*

This Proposition is the main result derived in the external demand imbalances model of Farhi and Werning (2014). I prove this result holds true also in a dynamic setting (their model is fully static) in Appendix A.2. The proof shows that, at a symmetric equilibrium, a decrease in regional population increases regional per-capita output,

$$\begin{aligned} \frac{\mathrm{d}y_{i,t}}{\mathrm{d}(-n_{i,t})} &> 0, \\ \frac{\mathrm{d}y_{i,t}}{\mathrm{d}(-n_{i,t})} &= \frac{y_{i,t} - g_{i,t}}{n_{i,t}}, \quad \text{if } \alpha_{N,t} = \alpha_{S,t} = \frac{1}{2} \end{aligned} \quad , \tag{29}$$

implying emigration has a spillover effect that increases per-capita output for the origin region. The derivation in Appendix A.2 shows the general expression of this derivative which is proportional to per-capita private demand for the region's final good (private consumption plus net exports). The case of no home bias ($\alpha_{N,t} = \alpha_{S,t} = 1/2$) shows this derivative in its simplest form. With home bias, this spillover effect will be smaller, but still positive under Keynesian-cross stability, $\partial c_{i,t}/\partial y_{i,t} < 1$, which I assume or impose parametrically.

Notably, the result does not originate from the production side which exhibits constant returns to scale. Instead, it relies on demand-side equations, the regional budget constraint and goods market clearing. The intuition behind this result is that, since both goods are traded with imperfect home bias, when agents migrate they continue to demand some of their origin region's goods from their new region of residence. Emigration thus reduces a region's population proportionally more than it reduces aggregate demand for that region's goods, raising per-capita output in the origin region. For this reason, the stronger home bias is, the weaker this spillover effect will be.

I interpret this spillover effect as being the Mundellian channel through which labour mobility—by migrating away from a bust region towards a boom region—reduces regional imbalances in a currency union.

PROPOSITION 2 (Equilibrium populations)**.** *Equilibrium populations are pinned down by the indifference condition of the marginal agent. Agents migrate towards the region with higher per-capita output.*

Under free mobility, agents choose their region of residence according to (9) to maximise lifetime utility. In equilibrium, the marginal agent is indifferent between living in the North (N) and the South (S). Indexing the marginal agent by a superscript *, lifetime utilities therefore satisfy

$$\begin{gathered} U^*(s^t|s_t = N) = U^*(s^t|s_t = S) \iff \\ \theta \log \frac{n_{N,t}}{n_{S,t}} = \log \frac{\tilde{c}_t^*(s^t|s_t = N)}{\tilde{c}_t^*(s^t|s_t = S)} + \log \frac{\xi_{N,t}}{\xi_{S,t}} + \beta \mathbb{E}_t\{U^*(s^{t+1}|s_t = N) - U^*(s^{t+1}|s_t = S)\}. \end{gathered} \tag{30}$$

By the budget constraint (3), the consumption and continuation terms differ across the two locations only through the regional wage. A first-order approximation of these terms in the wage gap $\log(W_{N,t}/W_{S,t})$, taken around a symmetric equilibrium, gives

$$\theta \log \frac{n_{N,t}}{n_{S,t}} \approx \frac{W_t \ell_t}{\tilde{c}_t^*} \log \frac{W_{N,t}}{W_{S,t}} + \log \frac{\xi_{N,t}}{\xi_{S,t}}. \tag{31}$$

By the envelope theorem, the induced changes in labour supply and asset holdings have no first-order effect. The same applies to the continuation values. Because agents can costlessly re-optimise their location next period, continuation values differ only through savings, and this difference also vanishes to first order. Intuitively, since profits are distributed uniformly across the union and agents take their assets with them when they migrate, wages are the only source of location-dependent income. Relative populations must therefore reflect relative wages, together with relative location preference shocks.

Regional wages follow from combining the labour market clearing condition (26) with labour supply (6) and labour demand (17):

$$\begin{gathered} W_{i,t} = \psi \left(\frac{y_{i,t}}{A}\right)^{\varphi} = \psi \left(\frac{Y_{i,t}}{A\, n_{i,t}}\right)^{\varphi} \implies \\ \log \frac{n_{N,t}}{n_{S,t}} = \log \frac{Y_{N,t}}{Y_{S,t}} - \frac{1}{\varphi} \log \frac{W_{N,t}}{W_{S,t}}. \end{gathered} \tag{32}$$

Combining (32) with the indifference condition (31) and the population constraint (10) gives, to a first-order approximation,

$$n_{i,t} \approx \frac{Y_{i,t}^{\kappa_t}\, \xi_{i,t}^{\frac{1-\kappa_t}{\theta}}}{Y_{N,t}^{\kappa_t}\, \xi_{N,t}^{\frac{1-\kappa_t}{\theta}} + Y_{S,t}^{\kappa_t}\, \xi_{S,t}^{\frac{1-\kappa_t}{\theta}}}, \tag{33}$$

where $\kappa_t \equiv \frac{\varphi}{\varphi + \theta \tilde{c}_t^*/(W_t \ell_t)}$ depends on the marginal agent's ratio of consumption to labour income at the symmetric equilibrium, and on the sensitivity of utility to congestion costs, θ.

To build intuition for (33), consider first the case with neither congestion costs nor preference shocks. Since wages are the only source of regional income differences, agents would then migrate until wages are equalised. This process converges because wages depend positively on per-capita output $y_{i,t}$, which, by the migration spillover of Proposition 1, falls as

population rises. Equilibrium populations would then be determined by (32) together with wage equalisation, in place of (31). Equal wages imply equal labour supply per worker, $\ell_{i,t}$, across regions, so labour market clearing requires the extensive margin—regional populations $n_{i,t}$—to adjust to labour demand. By (17), each region's population share would then equal its share of union output, $n_{i,t} = \frac{Y_{i,t}}{Y_{N,t}+Y_{S,t}}$. Congestion costs change this, since it is no longer wages that are equalised, but the lifetime utility of the marginal agent, which introduces the exponent $\kappa_t < 1$ on regional output. Preference shocks add a further, independent source of population differences, giving the final expression in (33).

Finally, log-differentiating the equilibrium population (33) around the symmetric equilibrium and in the absence of location preference shocks, and using the central bank's policy of keeping union-wide output constant, yields

$$\frac{\mathrm{d}n_{i,t}}{\mathrm{d}y_{i,t}} = \frac{\kappa_t}{1-\kappa_t}\frac{n_{i,t}}{y_{i,t}} > 0. \tag{34}$$

Agents therefore migrate towards the region with higher per-capita output.

PROPOSITION 3 (Effect of migration on fiscal space)**.** *Labour mobility exerts two opposing forces on fiscal space, having both a relaxing and a tightening effect on fiscal constraints. The sign of the net effect will mainly depend on the magnitude of the migration spillover effect, and on the existing debt servicing cost burdens.*

In any given period t, the stochastic debt limit $\bar{b}_{i,t}$ imposes an upper bound on how much the government can spend. Taking the government budget constraint (21), and using labour supply (6) and equilibrium wage (32), the government is fiscally constrained in its maximum period spending by

$$g_{i,t} \leq \psi\tau_{i,t}\left(\frac{y_{i,t}}{A}\right)^{1+\varphi} - \frac{1+r_{t-1}}{1+\gamma_{i,t}}b_{i,t-1} + \bar{b}_{i,t} \equiv \mathcal{F}_{i,t}, \tag{35}$$

where $\gamma_{i,t} \equiv \frac{n_{i,t}}{n_{i,t-1}} - 1$ denotes the net population growth rate. The right-hand side of the above fiscal constraint measures the government's fiscal space, $\mathcal{F}_{i,t}$.

Totally differentiating $\mathcal{F}_{i,t}$ to assess how fiscal space is impacted by emigration ($\mathrm{d}(-n_{i,t}) > 0$) yields

$$\frac{\mathrm{d}\mathcal{F}_{i,t}}{\mathrm{d}(-n_{i,t})} = \underbrace{(1+\varphi)\frac{\tau_{i,t}W_{i,t}\ell_{i,t}}{y_{i,t}}\cdot\frac{\mathrm{d}y_{i,t}}{\mathrm{d}(-n_{i,t})}}_{\text{Fiscal contribution channel}} - \underbrace{\frac{(1+r_{t-1})b_{i,t-1}}{(1+\gamma_{i,t})n_{i,t}}}_{\text{Debt burden channel}} \gtreqless 0. \tag{36}$$

When workers migrate out of the depressed region, the first term on the right-hand side is positive—as established by Proposition 1, per-capita output rises with emigration ($\mathrm{d}y_{i,t}/\mathrm{d}(-n_{i,t}) > 0$). Since per-capita output is rising as workers leave the region, per-capita tax revenue rises with it, expanding the government's fiscal space. This effect is naturally made stronger when the magnitude of the migration spillover on per-capita output is larger

(which as noted in Proposition 1 occurs for larger values of private demand), as this is the force that drives the positive fiscal contribution channel.[2]

The second term on the right-hand side indicates that there is a debt burden channel through which emigration contracts fiscal space. This captures the fact that, as workers migrate out of the region, the existing debt stock is spread over fewer workers, increasing the per-capita debt burden and hence shrinking the government's fiscal space. Intuitively, even if government spending scales one-for-one with population, the existing debt stock is incapable of such scaling as it needs to be repaid in full and in aggregate terms. Emigration leads to there being fewer workers to repay this debt, and the per-capita burden of debt repayment increases with emigration. More generally, notice that how the debt burden channel evolves depends on the sign of $r_{t-1} - \gamma_{i,t}$, a condition analogous to the classic $r - g$ determinant of debt sustainability in the public finance literature (Domar, 1944; Blanchard et al., 1991; Blanchard, 2019; Mehrotra and Sergeyev, 2021), here with population growth replacing GDP growth. Emigration shrinks the population ($\gamma_{i,t} < 0$) which, for a given interest rate on sovereign debt r_{t-1}, makes the per-capita debt burden grow faster, contracting the government's fiscal space.

Equation (36) is sufficiently general, and the ambiguity of how fiscal space reacts to emigration holds even if we allow for other types of government expenses and transfers. Namely, if the government paid non-employment (as a proxy for unemployment) benefits $\omega_{i,t} W_{i,t}(1 - \ell_{i,t})$ with replacement rate $\omega_{i,t}$, this expense would also decline through the migration spillover, and the savings on these benefit transfers would simply reinforce the positive effect of emigration on fiscal space, under a more general *net* per-capita fiscal contribution channel. Similarly, if the government were not able to fully scale its consumption proportionally with population (implying government consumption per capita was $g_{i,t} + G_i/n_{i,t}$), any fixed non-scalable component of government consumption (e.g. administrative expenses, infrastructure maintenance) $G_i/n_{i,t}$ would reinforce the negative effect of emigration on fiscal space, through a more general non-scalable expenses burden channel.

III.B Fiscal-mobility multipliers

On impact, with prices pre-determined, regional output is determined by regional demand through the goods market clearing condition (25). Equilibrium populations are determined to first order by condition (33), derived in Proposition 2. Government spending—and fiscal policy more generally—is given by the minimum of two levels: the level prescribed by the fiscal rule (20), and the government's fiscal space (35), as defined in Proposition 3.

[2] It is also stronger when the elasticity of tax revenue with respect to output—which equals the $(1 + \varphi)\frac{\tau_{i,t} W_{i,t} \ell_{i,t}}{y_{i,t}}$ term—is higher, which occurs for lower values of the Frisch elasticity (higher φ). Intuitively, under sticky prices, firms hire whatever labour is required to meet demand, with the per-capita labour hour requirement rising with the emigration-induced higher $y_{i,t}$. A lower Frisch elasticity means a steeper wage hike is required to increase labour hours supplied, amplifying the wage bill and hence per-worker tax revenue. The corresponding fall in firm profits is not taxed (besides being borne by all union households under global profit sharing), so does not offset the regional fiscal gain.

Together, these conditions form a system of equations linking three key variables: per-capita output $y_{i,t}$, per-capita government spending $g_{i,t}$ and population $n_{i,t}$. A convenient point of approximation is a symmetric equilibrium with no home bias. Log-linearising around this point yields three equations: a regional demand equation (RD); a labour mobility condition (LM); and a fiscal policy equation, which follows either the counter-cyclical fiscal rule (FR) when the fiscal constraint is slack, or the pro-cyclical fiscal space (FS) when it binds. Throughout this section, I hold fiscal policy in the other region fixed, $\hat{g}_{-i,t} = 0$. The three equations are

$$\hat{y}_{i,t} = \frac{1}{2}\frac{g_t}{y_t}\,\hat{g}_{i,t} - \frac{c_t}{y_t}\,\hat{n}_{i,t} + \varepsilon^{RD}_{i,t}, \tag{RD}$$

$$\hat{n}_{i,t} = \frac{\kappa_t}{1-\kappa_t}\,\hat{y}_{i,t} + \varepsilon^{LM}_{i,t}, \tag{LM}$$

$$\hat{g}_{i,t} = \begin{cases} -\rho_g \dfrac{y_t}{g_t}\hat{y}_{i,t} + \varepsilon^{FR}_{i,t}, & \text{if } g^{\star}_{i,t} \leq \mathcal{F}_{i,t} \quad \text{(FR)} \\ (1+\varphi)\dfrac{\tau_t W_t \ell_t}{g_t}\,\hat{y}_{i,t} + \dfrac{(1+r_{t-1})b_{i,t-1}}{g_t}\,\hat{n}_{i,t} + \varepsilon^{FS}_{i,t}, & \text{if } g^{\star}_{i,t} > \mathcal{F}_{i,t} \quad \text{(FS)} \end{cases}, \tag{FR/FS}$$

where $\hat{x} \equiv \mathrm{d}x/x_t$ for any variable x, and $g^{\star}_{i,t} \equiv g_0 - \rho_g(y_{i,t} - y_0) + \zeta^g_{i,t}$ denotes the unconstrained, desired level of counter-cyclical government spending. Each equation is subject to its own shock. Regional demand shocks arise from stochastic changes in home bias, $\varepsilon^{RD}_{i,t} = \frac{c_t}{y_t}\mathrm{d}(\alpha_{i,t} - \alpha_{-i,t})$. Labour mobility shocks arise from location preference shocks, $\varepsilon^{LM}_{i,t} = \frac{1}{2\theta}(\hat{\xi}_{i,t} - \hat{\xi}_{-i,t})$. The fiscal rule shock is $\varepsilon^{FR}_{i,t} = \frac{\mathrm{d}\zeta^g_{i,t}}{g_t}$, and fiscal space shocks arise from exogenous changes in the real per-capita debt limit, $\varepsilon^{FS}_{i,t} = \frac{\mathrm{d}\bar{b}_{i,t}}{g_t}$.[3]

The coefficients have direct interpretations. In (RD), the positive coefficient on $\hat{g}_{i,t}$ is the partial-equilibrium fiscal multiplier, while the negative coefficient on $\hat{n}_{i,t}$ captures the migration spillover of Proposition 1 and equals the share of private consumption in output. In (LM), population responds positively to output, with a slope governed by κ_t, defined in Proposition 2, and which is decreasing in θ, the sensitivity of utility to congestion costs. The negative slope in (FR) reflects the counter-cyclical fiscal rule. The coefficients in (FS) show that fiscal space expands when output rises, through higher tax revenue, and contracts when population falls, as legacy debt is spread over fewer taxpayers.

Two features of the approximation deserve comment. First, although the log-linearisation is taken around a symmetric equilibrium, it does not require legacy debts $b_{i,t-1}$ to be symmetric. I therefore keep $b_{i,t-1}$ region-specific, which allows the debt burden channel to carry different weights across regions. Second, the no-home-bias assumption ($\alpha_{N,t} = \alpha_{S,t} = 1/2$) is what allows the consumption responses to drop out of the system. Since the central bank

[3] In order to rule out self-financing spending, I assume throughout that $(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t} < 2$. A unit increase in government spending raises output by $\frac{1}{2}$, and the assumption ensures that the resulting tax revenue falls short of the original unit of spending. The feedback loop between $\hat{y}_{i,t}$ and $\hat{g}_{i,t}$ in (RD) and (FS) is therefore non-explosive. When the debt burden exceeds $2\frac{c_t}{y_t}$, the feedback loop through migration must also be non-explosive, which requires $\frac{\kappa_t}{1-\kappa_t}\left(\frac{1}{2}\frac{(1+r_{t-1})b_{i,t-1}}{y_t} - \frac{c_t}{y_t}\right) < 1 - \frac{1}{2}(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}$, and I assume this also holds throughout.

holds union-wide output fixed, *regional* fiscal policy is fully crowded out at the *union* level: when populations are symmetric, $\mathrm{d}c_{i,t} + \mathrm{d}c_{-i,t} = -\mathrm{d}g_{i,t}$. With no home bias, this union-wide fall in consumption falls equally on both regions' output, so the relative sizes of $\mathrm{d}c_{i,t}$ and $\mathrm{d}c_{-i,t}$ do not matter. This is the source of the $1/2$ coefficient on $\hat{g}_{i,t}$ in (RD). Similarly, the coefficient on $\hat{n}_{i,t}$ in (RD) can only be obtained ignoring consumption responses under no home bias. The no-home-bias point is therefore not intended as a realistic description of the economy, but as a point around which the system can be solved in a tractable manner and independent of consumption responses. I show in Appendices A.2 and A.3, respectively, that the coefficients on $\hat{n}_{i,t}$ and $\hat{g}_{i,t}$ in (RD) keep their signs in the presence of home bias, so all results below hold qualitatively. Quantitatively, home bias makes the coefficient on $\hat{g}_{i,t}$ slightly larger and the coefficient on $\hat{n}_{i,t}$ less negative. Moreover, although the approximation point has no home bias, the regional demand shocks $\varepsilon^{RD}_{i,t}$ are driven precisely by changes in the α's, so the analysis still allows for shocks that make regions home-biased.

Under this approximation, the result is a linear system of three equations in three unknowns, $(\hat{y}_{i,t}, \hat{n}_{i,t}, \hat{g}_{i,t})$, which can be solved in closed form. The full solution to the system can be found in Appendix A.4.

Unconstrained government. To establish a benchmark as to how shocks propagate in the economy, we can fully solve the above system of equations for an unconstrained government whose fiscal policy follows its counter-cyclical fiscal rule (FR). In this case, solving for $\hat{y}_{i,t}$, we obtain

$$\hat{y}_{i,t} = \mathcal{M}^{FR}_t \cdot \left(\varepsilon^{RD}_{i,t} + \frac{1}{2}\frac{g_t}{y_t}\,\varepsilon^{FR}_{i,t} - \frac{c_t}{y_t}\,\varepsilon^{LM}_{i,t} \right),$$

$$\mathcal{M}^{FR}_t = \frac{1}{1 + \underbrace{\tfrac{1}{2}\rho_g}_{\substack{\text{Fiscal}\\ \text{stabilisation}}} + \underbrace{\tfrac{\kappa_t}{1-\kappa_t}\tfrac{c_t}{y_t}}_{\substack{\text{Mundellian}\\ \text{stabilisation}}}} < 1. \tag{37}$$

The multiplier $\mathcal{M}^{FR}_t$ scales every shock to the regional economy ($\varepsilon^{RD}_{i,t}$, $\varepsilon^{LM}_{i,t}$, $\varepsilon^{FR}_{i,t}$). It is strictly less than one, meaning that in general equilibrium, a unit shock to regional demand moves per-capita output by less than one-for-one. Two stabilising forces are responsible. The first is counter-cyclical fiscal policy. In a boom, the government cuts spending by ρ_g for each unit increase in output, and each unit cut lowers regional output by $1/2$. The second is the Mundellian mechanism. Higher (lower) regional demand raises (lowers) regional population with elasticity $\frac{\kappa_t}{1-\kappa_t}$, which in turn lowers (raises) per-capita demand by a factor of $\frac{c_t}{y_t}$. Per-capita output therefore moves by less than the full amount of the original shock.

These two mechanisms reflect how fiscal policy and labour mobility have been viewed in the optimal currency area literature to date—both are stabilising, but they operate separately and do not interact. This is indeed the case when governments are fiscally unconstrained. It is no longer the case, however, once fiscal constraints bind, because fiscal policy and labour mobility then interact through fiscal space.

Constrained government with no labour mobility. Now consider a fiscally constrained government, whose spending is dictated by the pro-cyclical fiscal space (FS). Fiscal constraints affect shock propagation independently of any interaction with labour mobility. To isolate this effect, I first shut down labour mobility by imposing $\hat{n}_{i,t} = 0$, and reintroduce it below. Solving for $\hat{y}_{i,t}$ gives

$$\begin{aligned} \hat{y}_{i,t}|_{\hat{n}_{i,t}=0} &= \mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0} \cdot \left(\varepsilon_{i,t}^{RD} + \frac{1}{2}\frac{g_t}{y_t}\,\varepsilon_{i,t}^{FS}\right), \\ \mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0} &= \frac{1}{1 - \underbrace{\frac{1}{2}(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}}_{\text{Forced consolidation}}} > 1. \end{aligned} \tag{38}$$

The multiplier $\mathcal{M}_t^{FS}|_{\hat{n}_{i,t}=0}$ is now strictly greater than one. Idiosyncratic regional shocks are amplified rather than stabilised. Without labour mobility there is, naturally, no Mundellian stabilisation. Instead, a negative demand shock lowers per-capita output and hence tax revenue, with revenue elasticity $(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}$. Because fiscal space binds, government spending must fall by this amount, and this pro-cyclical cut feeds back into output with a coefficient of $1/2$. This is the familiar result that forced austerity during recessions amplifies the shock that caused the downturn. The question is whether labour mobility—through Mundellian stabilisation and through its effect on fiscal space—can offset this amplification and cushion the forced consolidation.

Constrained government with labour mobility. Allowing population to adjust according to (LM) reintroduces Mundellian stabilisation, but it also activates the two channels through which labour mobility affects fiscal space: the fiscal contribution channel and the debt burden channel. Depending on their relative magnitudes, these channels may ease or reinforce the forced consolidation. With labour mobility, the solution for a fiscally constrained government becomes

$$\hat{y}_{i,t} = \mathcal{M}_t^{FS} \cdot \left(\varepsilon_{i,t}^{RD} + \frac{1}{2}\frac{g_t}{y_t}\,\varepsilon_{i,t}^{FS} - \left[\frac{c_t}{y_t} - \frac{1}{2}\frac{(1+r_{t-1})b_{i,t-1}}{y_t}\right]\varepsilon_{i,t}^{LM}\right),$$

$$\begin{aligned} \frac{\mathcal{M}_t^{FS}}{\mathcal{M}_t^{FS}|_{\hat{n}_{i,t}=0}} = \Bigg[1 + \underbrace{\frac{\kappa_t}{1-\kappa_t}\frac{c_t}{y_t}}_{\substack{\text{Mundellian}\\ \text{stabilisation}}} + \frac{1}{2}\,\mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0}\underbrace{(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}\frac{\kappa_t}{1-\kappa_t}\frac{c_t}{y_t}}_{\substack{\text{Fiscal contribution}\\ \text{channel stabilisation}}} \\ - \frac{1}{2}\,\mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0}\underbrace{\frac{(1+r_{t-1})\,b_{i,t-1}}{y_t}\frac{\kappa_t}{1-\kappa_t}}_{\substack{\text{Debt burden}\\ \text{channel amplification}}}\Bigg]^{-1} \gtreqless 1. \end{aligned} \tag{39}$$

Labour mobility can therefore either raise or lower the multiplier of a fiscally constrained region. It does so through two effects: the Mundellian effect and the fiscal-mobility interaction.

The latter combines the fiscal contribution and debt burden channels of Proposition 3, which pull in opposite directions. The fiscal contribution channel reinforces Mundellian stabilisation and lowers the multiplier, whereas the debt burden channel further amplifies idiosyncratic shocks and regional imbalances.

Each term in the expression above has a direct interpretation. The Mundellian stabilisation term is identical to that in (37). The fiscal contribution channel is the product of four factors: (i) the Mundellian cushioning of output, $\frac{\kappa_t}{1-\kappa_t}\frac{c_t}{y_t}$; (ii) the tax revenue elasticity $(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}$, through which this cushioning raises revenue; (iii) the forced-consolidation amplification $\mathcal{M}_t^{FS}\big|_{\hat{n}_{i,t}=0}$, which the additional revenue relaxes; and (iv) the effect, $1/2$, of the resulting increase in government spending on output. The debt burden channel is likewise the product of four factors: (i) the population outflow per unit fall in output, $\frac{\kappa_t}{1-\kappa_t}$; (ii) the legacy debt burden $\frac{(1+r_{t-1})b_{i,t-1}}{y_t}$, which is now spread over fewer workers; (iii) the forced-consolidation amplification $\mathcal{M}_t^{FS}\big|_{\hat{n}_{i,t}=0}$, which the heavier per-capita burden reinforces; and (iv) again, the effect $1/2$ of spending on output.

The net effect of these two fiscal-mobility channels determines whether the interaction of labour mobility with fiscal space alleviates or reinforces the forced consolidation. Its sign and magnitude, in turn, determine whether the fiscal-mobility interaction works alongside Mundellian stabilisation or against it—potentially outweighing it altogether. These different scenarios give rise to different fiscal-mobility regimes.

III.C Fiscal-mobility regimes

The previous subsection shows that the interaction between labour mobility and fiscal space can either relax or tighten fiscal constraints, and can therefore either reinforce Mundellian stabilisation or work against it, potentially overturning it altogether. This raises the question of when each of these scenarios occurs. Both the Mundellian effect and the fiscal contribution channel are stronger when the migration spillover, $\frac{c_t}{y_t}$, is large. The debt burden channel, by contrast, is stronger when the legacy debt burden, $\frac{(1+r_{t-1})b_{i,t-1}}{y_t}$, is high. It is therefore natural to characterise the different regimes by comparing the strength of the migration spillover with the size of legacy debt.

In the first regime, the fiscal contribution channel dominates the debt burden channel, so labour mobility has a net positive effect on fiscal space. Fiscal constraints are relaxed, consolidations are milder than they would be without labour mobility, and the fiscal-mobility interaction aids Mundellian stabilisation. I call this regime *fiscal-mobility stabilisation*. It occurs whenever

$$\underbrace{(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}\frac{c_t}{y_t}}_{\text{Fiscal contribution channel}} \geq \underbrace{\frac{(1+r_{t-1})b_{i,t-1}}{y_t}}_{\text{Debt burden channel}}. \tag{40}$$

In the second regime, the debt burden channel dominates the fiscal contribution channel. Emigration then has a net negative effect on fiscal space, forcing deeper consolidations than without labour mobility, and the fiscal-mobility interaction works against Mundellian

stabilisation. The final outcome depends on whether fiscal-mobility amplification is weaker or stronger than Mundellian stabilisation. When it is weaker, consolidations are more severe, but the fall in output is still cushioned relative to the case without labour mobility. I call this regime *partial fiscal-mobility amplification*. This occurs whenever

$$\underbrace{(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}\frac{c_t}{y_t}}_{\text{Fiscal contribution channel}} < \underbrace{\frac{(1+r_{t-1})b_{i,t-1}}{y_t}}_{\text{Debt burden channel}} \leq \underbrace{2\frac{c_t}{y_t}}_{\text{Mundell}} . \tag{41}$$

The factor of 2 on the Mundellian term arises because changes in fiscal space affect $\hat{y}_{i,t}$ only through $\hat{g}_{i,t}$, with a coefficient of $\frac{1}{2}$, whereas the Mundellian effect acts on $\hat{y}_{i,t}$ directly. The interval above is non-empty whenever $(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t} < 2$, which is exactly the assumption imposed earlier to rule out self-financing spending and explosive feedback between $\hat{y}_{i,t}$ and $\hat{g}_{i,t}$.

When fiscal-mobility amplification is instead stronger than Mundellian stabilisation, output falls by more than it would without labour mobility. A region in a bust would then be better off without labour mobility. I call this regime *full fiscal-mobility amplification*. This occurs whenever

$$\underbrace{\frac{(1+r_{t-1})b_{i,t-1}}{y_t}}_{\text{Debt burden channel}} > \underbrace{2\frac{c_t}{y_t}}_{\text{Mundell}} . \tag{42}$$

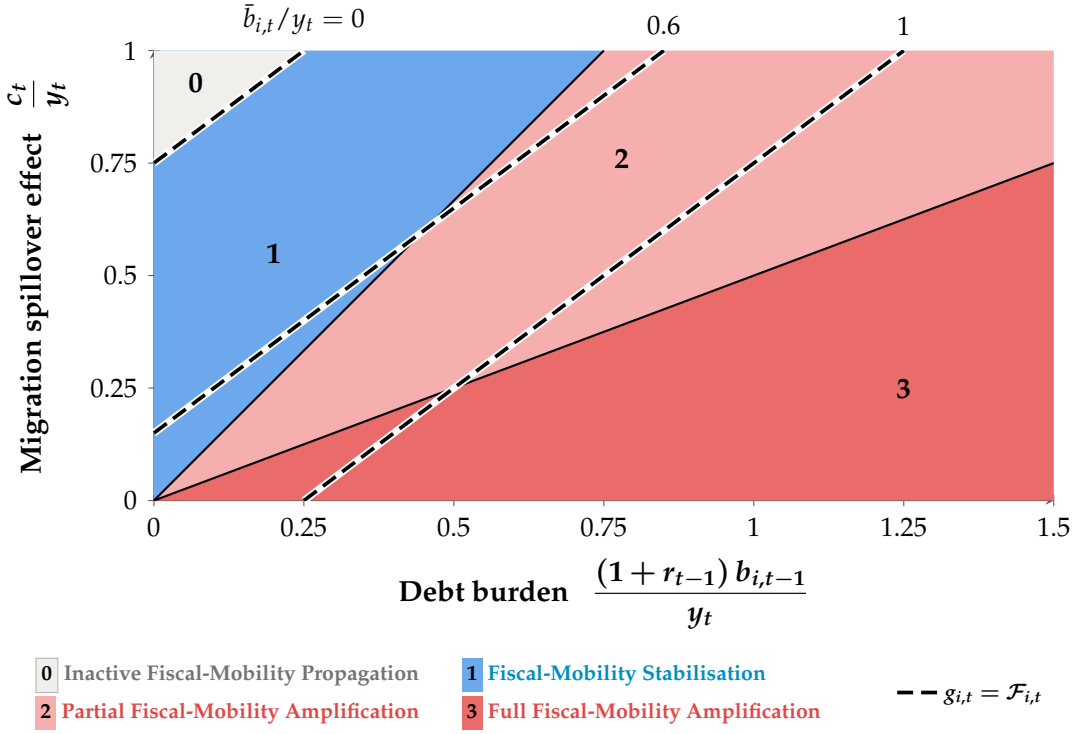
Notably, the fiscal contribution channel does not enter the threshold $2\frac{c_t}{y_t}$ separating these two cases. This is because the fiscal contribution channel is itself a by-product of Mundellian stabilisation: it is the additional tax revenue generated by the output that Mundellian stabilisation preserves, amplified by the forced-consolidation multiplier $\mathcal{M}_t^{FS}\big|_{\hat{n}_{i,t}=0}$. The Mundellian effect and the fiscal contribution channel together therefore amount to the Mundellian effect scaled by $\mathcal{M}_t^{FS}\big|_{\hat{n}_{i,t}=0}$. Since the debt burden channel is scaled by the same multiplier, it cancels from the comparison, leaving only the debt burden and the migration spillover. Figure 2 plots the different regimes against the legacy debt burden and the migration spillover, for an illustrative parameter configuration.

An important caveat, however, is that all regimes require fiscal constraints to bind, implying that government consumption tracks fiscal space, $g_{i,t} = \mathcal{F}_{i,t}$. This condition is equivalent to

$$\underbrace{\frac{(1+r_{t-1})b_{i,t-1}}{y_t}}_{\text{Debt burden}} = \underbrace{\frac{c_t}{y_t}}_{\substack{\text{Migration}\\ \text{spillover}}} + \frac{\tau_t W_t \ell_t}{y_t} + \frac{\bar{b}_{i,t}}{y_t} - 1, \tag{43}$$

which crucially implies that the debt burden and the migration spillover are not independent of each other at the binding fiscal constraint. Whenever the left-hand side of the above expression is smaller than the right-hand side, fiscal constraints are slack and we are in an *inactive fiscal-mobility propagation* region. Legacy debt therefore determines the strength of the debt burden channel but, through (43), it will also determine the fiscal-mobility regime that a

FIGURE 2: Mapping of Fiscal-Mobility Regimes



Notes: Illustrative values $(1+\varphi)\,\tau_t W_t \ell_t / y_t = 0.75$ and $\varphi = 2$. Coloured areas show the regime of a region whose fiscal constraint binds. Each dashed line is (43) with equality for the debt limit $\bar{b}_{i,t}/y_t$ indicated. A region with that limit is slack to the left of its line and binding on it, and a looser limit shifts the line to the right one for one. In the grey area the constraint cannot bind for any non-negative debt limit.

constrained government falls in.

The analytical results of this section yield several predictions that can be tested in the data. (i) Emigration has opposing effects on fiscal space, raising the net per-capita fiscal contributions of those who remain, while increasing the per-capita burden of legacy debt and other non-scalable expenses. (ii) Unconstrained governments raise spending in response to negative idiosyncratic shocks, stabilising regional imbalances. (iii) Constrained governments are instead forced into consolidations, whose extent depends on the net effect of emigration on fiscal space; depending on the sign and magnitude of this net effect, fiscal-mobility interactions may either stabilise or amplify regional shocks. (iv) For a constrained government, emigration deepens consolidations whenever the debt burden channel dominates the fiscal contribution channel, which for a given migration spillover is more likely the higher the legacy debt. I now turn to testing some of these predictions empirically.

IV Data

In order to empirically test some of the predictions of the model, I compile a monthly panel dataset on intra-EU migration, public finance, and economic activity statistics for the 28 countries that have been member states of the European Union in the 21st century. My data begins in January 2000, a year after the introduction of the euro and when many of the

data series begin. I end my sample period in December 2019 to exclude the Covid shock, which had very particular consequences on public finances and labour mobility, which may significantly differ from the mechanisms I try to capture. The choice of including all EU countries as opposed to Euro Area countries is due to the fact that monetary policy plays a minimal role in the predictions of Section III, and the key institutional features of the model—legal freedom of movement of labour and autonomous regional fiscal policy—are characteristics of the EU, and not of the Euro Area.

Migration flows and rates. Empirical studies on migration have often been constrained by data availability. Official national statistics are severely lacking in terms of granularity, and are often very incomplete. In terms of micro data, there is very little official information on who migrates, and consequently applied micro studies often have to rely on data for very niche movers, such as footballers, inventors, or millionaires (see Kleven et al. 2020 for a review). Aggregate data is equally very incomplete, only available at yearly frequencies, and varying a lot in quality—many countries do not record all bilateral flows, a problem which is exacerbated within the Schengen Area. Even among countries that report bilateral flows, emigration is well known to be much more poorly measured than immigration—migrants have an incentive to register their residence in their receiving country to benefit from public services, while not having an incentive to report their change of residence to their origin country. Furthermore, aggregate data does not capture many types of migration flows, such as seasonal or circular migration.

As a solution to these limitations, I use data from Revelio Labs which collects information on LinkedIn job positions and education. Other papers have previously used similar approaches of constructing bilateral migration measures from social media accounts, such as Zagheni et al. (2017) and Chi et al. (2025) who use Facebook users' updates to their country of residence. LinkedIn has the additional advantage that it is more directly related to the labour market, and users often update their CV retroactively, allowing for coverage of earlier periods. Through this, I construct a detailed monthly series of self-reported work- and education-related mobility within the EU. Besides being reported at monthly rather than yearly frequency, other advantages include capturing migration patterns left out by official statistics, such as seasonal and circular migration, as well as capturing intra-Schengen migration which is not recorded by many national authorities. I record a migration *flow* whenever a LinkedIn user's CV states they started a new job or education in an EU country, where their chronologically previous position was in a different EU country. The data is highly seasonal, with migration peaks in January and after the summer period.[4] To deal with this, I apply an X-13 seasonal adjustment filter directly to each country's immigration and

[4]A concern is whether this is true seasonality, or whether it is a data artefact where users do not fill out the month field for each job entry on their profile, in which case Revelio Labs codes the month to January. I verify that less than 5% of completed job positions involving a migration have both start and end month coded to January, suggesting that most users fill out the month field. Revelio Labs does not report the month for education, and therefore I hard-code the start month to September, the most likely month for starting a degree. The seasonal adjustment mitigates measurement error concerns from this approach.

emigration series, aggregated from users' sample weights. Aggregate migration *rates* are then computed as a share of the cubic trend of reported LinkedIn active employment or education, which ensures migration rates are scaled by the growth in the platform's user base. More details on data cleaning can be found in Appendix A.5.1.

There are, nonetheless, also many concerns on the validity of migration flows implied by LinkedIn. Not everyone has a LinkedIn account, not everyone who has an account might update it regularly or accurately,[5] and those who do have an account are highly selected into white-collar jobs. To mitigate these concerns, I use the sample weights calculated by Revelio Labs to take into account the probability of each user having an account based on their occupation and geography. I furthermore perform some validation exercises in Appendix A.6. First, Figure A.1 plots annual bilateral migration flows implied from aggregating the LinkedIn series against available national data,[6] showing a strong log-log correlation of 0.83, and a slope coefficient statistically indistinguishable from 1. Second, Figure A.2 plots the time series of the net migration rate for each country, constructed from the monthly LinkedIn series and the annual official statistics, again showing a very strong overlap between the two series for most countries.[7]

Other variables and data sources. Another key variable that is difficult to obtain at a monthly frequency is government debt stock, which is mostly reported on a quarterly basis. However, I construct monthly debt stock series by exploiting the fact that debt securities—the largest instrument of total government debt—are widely available at monthly frequencies. The main data source used for this procedure is the ECB's and Eurostat's Government Finance Statistics dataset, complemented with the ECB's alternative CSDB-derived Securities Issues Statistics to cover some missing data gaps. This data reveals that debt securities, reported at a monthly frequency, constitute on average 72% of total government debt, and are available for 75% of observations. This implies that roughly 55% of monthly variation in sovereign debt stocks can be observed and constructed directly. The variation in the remaining debt instruments (loans, and currency and deposits) is linearly interpolated from quarterly to monthly, and the same is done for the missing observations of debt securities. Since debt is a slow-moving stock variable, it is unlikely that there will be a lot of deviation from this linear interpolation over a three-month window, minimising concerns of measurement error. Furthermore, since I am mostly interested in variations in debt level rather than changes in output, I construct debt-to-GDP ratios as the ratio between the monthly debt series and *trend* GDP. Trend GDP is estimated by regressing quarterly log GDP on country-specific cubic time

[5]The sample starts in January 2000 but LinkedIn only publicly launched its platform in May 2003. Moreover, many users may only have created an account later in the sample. This is not a concern as long as users update their profiles retroactively to include job positions existing prior to the creation of their account. Moreover, while the lower usage of the platform in early years may affect migration flows, it should not affect migration rates, which are scaled by platform size.

[6]For 28 EU countries and a 20-year period between 2000 and 2019, there exists a total of 15,120 origin-destination-year flow triads. While from LinkedIn I get data for all these observations, national statistics only have data for 8,178 of these flows, demonstrating how poorly migration flows within Schengen are reported.

[7]Appendix A.6 discusses further possible reasons for existing discrepancies in more detail.

trends and exponentiating the monthly fitted values. A more detailed description of how the monthly debt stock series was constructed and treated, namely data consolidation, is included in Appendix A.5.2.

Additional variables used in the regression analysis include monthly unemployment rates, industrial production, and government bond yields. Unemployment rates are taken from the monthly Eurostat seasonally adjusted series for all EU-28 countries.[8] Monthly industrial production is also taken from Eurostat's seasonally and calendar adjusted data, indexed at 2015 volume levels. It covers production in mining and quarrying, manufacturing, electricity, gas, steam, and air conditioning supply. Finally, government bond yields are also sourced from Eurostat, using its long-term interest rate series based on the Maastricht Treaty's EMU convergence criterion, which tracks central government bond yields at a residual maturity of around 10 years. This variable is the binding impediment in obtaining a fully balanced panel, as some countries do not have 10-year government bonds over the sample period. Instead of using bonds with different maturities that would reduce comparability between countries, I opt to drop these observations, resulting in an unbalanced panel. The detailed list of missing observations is included in Appendix A.5.3.

Overall, in the main panel dataset, I have monthly observations on 28 countries for most of the 240-month period ranging from January 2000 to December 2019. Due to the unavailability of 10-year government bonds for some months and countries, the final dataset is an unbalanced panel of 6,196 observations in total. The empirical section also estimates the response of fiscal variables which are only available at quarterly frequency. For this purpose, government consumption was also obtained for the same panel of EU-28 countries between 2000 Q1 and 2019 Q4, sourced from Eurostat.

V Empirical Evidence

I use the monthly panel of EU countries between January 2000 and December 2019 to empirically test the predictions of Section III. The EU during this period is a natural laboratory for fiscal-mobility interactions. Most member states shared a monetary authority and therefore relied on labour and fiscal adjustments to absorb cross-country imbalances. The EU allows for free movement of labour, but mobility is too low (Arpaia et al., 2016) to dispense with the need for fiscal stabilisation. Furthermore, there is no federal fiscal authority, so each member state must rely on its own fiscal capacity. During the sovereign debt crisis, many countries breached the institutionally imposed 60% Maastricht debt ceiling, and several reached the market-implied debt limits. Fiscal constraints were therefore binding, and with workers emigrating *en masse* from depressed peripheral economies, the EU provides an ideal setting to test fiscal-mobility interactions.

[8] Estonia is missing one observation, which is filled in with data from the OECD.

V.A SVAR model

Identification. I estimate a structural vector autoregression (SVAR) model on the monthly panel of EU countries. The main goal of the SVAR is to obtain identified structural migration shocks, while already establishing some causal evidence between variables at a monthly frequency. The SVAR specification estimates a system of equations with 36 monthly lags, according to

$$\mathbf{Y}_{i,t} = \sum_{k=1}^{K=36} \mathbf{A}_k \mathbf{Y}_{i,t-k} + \alpha_i + \delta_{m(i,t),t} + \mathbf{B}\boldsymbol{\varepsilon}_{i,t}. \tag{44}$$

$\mathbf{Y}_{i,t}$ is a vector of endogenous variables on migration, fiscal space, and economic activity for country i in month t. The matrix $\mathbf{A}_k$ encodes own- and cross-variable effects, from the kth lag to the contemporaneous values. Matrix $\mathbf{B}$ maps the vector of orthogonal i.i.d. structural shocks $\boldsymbol{\varepsilon}_{i,t}$ to the contemporaneous responses of the endogenous variables. It is assumed that all matrices $\mathbf{A}_k$ and $\mathbf{B}$ are invariant across time and countries, and that there is no cross-dependence of endogenous variables across countries, i.e. a country's endogenous variables respond to their own lags, but not the lags of others.

The vector of endogenous variables $\mathbf{Y}_{i,t}$ includes the monthly immigration and emigration rates constructed from the LinkedIn data (both annualised), the debt-to-trend-GDP ratio constructed from the monthly debt securities series, the unemployment rate, the natural logarithm of an industrial production index, and the harmonised 10-year government bond yield. The system is estimated by OLS, with the inclusion of country fixed effects α_i,[9] and time fixed effects $\delta_{m(i,t),t}$. Since not all EU member states are part of the Euro Area (and hence maintain monetary autonomy), and since some countries in my panel joined the EU mid-sample, the time fixed effect is allowed to differ according to each country's EU and Euro Area (EA) membership status at a given point in time, $m(i,t) \in \{\text{non-EU}; \text{EU}; \text{EA}\}$. This flexibility accounts for exposure to common Euro Area monetary policy, or other EU policies commonly affecting migration and fiscal space.

OLS estimation of the reduced-form VAR does not separately identify the vector of structural shocks $\boldsymbol{\varepsilon}_{i,t}$ and its mapping matrix $\mathbf{B}$. Instead it only recovers their product $\mathbf{B}\boldsymbol{\varepsilon}_{i,t}$, the reduced-form residuals. As is standard in SVAR estimation, identification requires additional restrictions. I impose short-run timing restrictions (Sims, 1980) via a Cholesky decomposition of the reduced-form variance-covariance matrix, with variables ordered as: immigration rate, emigration rate, debt-to-trend-GDP, log industrial production, unemployment, bond yield. I follow the timing logic of Blanchard and Perotti (2002) in treating slow-moving fiscal variables (debt in my case) as pre-determined relative to real activity, and Favero and Giavazzi (2007) in ordering fast-moving financial variables (bond yields) last. The identifying

[9] The inclusion of country fixed effects in the dynamic SVAR model introduces Nickell (1981) bias in estimates. However, given that I have a long panel (average $\bar{T}$ for each country is 229, for a maximum T of 240 between January 2000 and December 2019), the bias is of order $O(1/T)$, and with 36 lags each country has an average effective sample of $\bar{T} - K = 229 - 36 = 193$ observations (see Alvarez and Arellano 2003 for the formal result that the within-groups estimator is consistent as $T \to \infty$). For this reason, I ignore possible Nickell biases and include country fixed effects to avoid model misspecification.

assumption is that migration does not respond within the same month to shocks to debt, real activity, or bond yields. I order immigration before emigration on the assumption that workers residing outside a country respond more slowly to its economic conditions than workers already residing within it, who can observe and react to local conditions directly. Results are robust to reversing this ordering between migration rates.

The main advantage that the SVAR approach offers is that it achieves identification without resorting to external instruments. Finding monthly, country-heterogeneous instruments for both fiscal space and labour mobility is impractical.[10] National policy shocks apply only to the country in question, while EU-wide shocks are scarce and their differential impact across countries is difficult to identify exogenously. The SVAR identification circumvents this problem by relying instead on short-run timing restrictions, as done by the literature when instruments for a large panel of countries are unavailable (Ilzetzki et al., 2013).

This requirement highlights the importance of constructing a higher-frequency monthly panel, as opposed to the more readily available lower-frequency annual data on migration and debt. Since identification relies on the short-run timing restrictions rather than some exogenous variation in an instrument, the meaning of "contemporaneous effects" across variables is highly sensitive to the frequency of the data. The identifying assumption that migration does not respond contemporaneously to any fiscal space or economic activity shock is particularly plausible if we are talking about same-*month* responses. The decision to migrate is a costly one, both financially and personally, and it is believable that families do not relocate across national borders within the same month that fiscal space contracts or economic activity drops. The same assumption is less believable if contemporaneous response means same-*year* response. If a persistent shock that contracts fiscal space or leaves families unemployed hits early in the year, it is more plausible that workers would choose to emigrate by the end of the year, violating the identifying assumption under annual frequency but not monthly frequency. Monthly frequency is therefore a key feature of my structural identification.

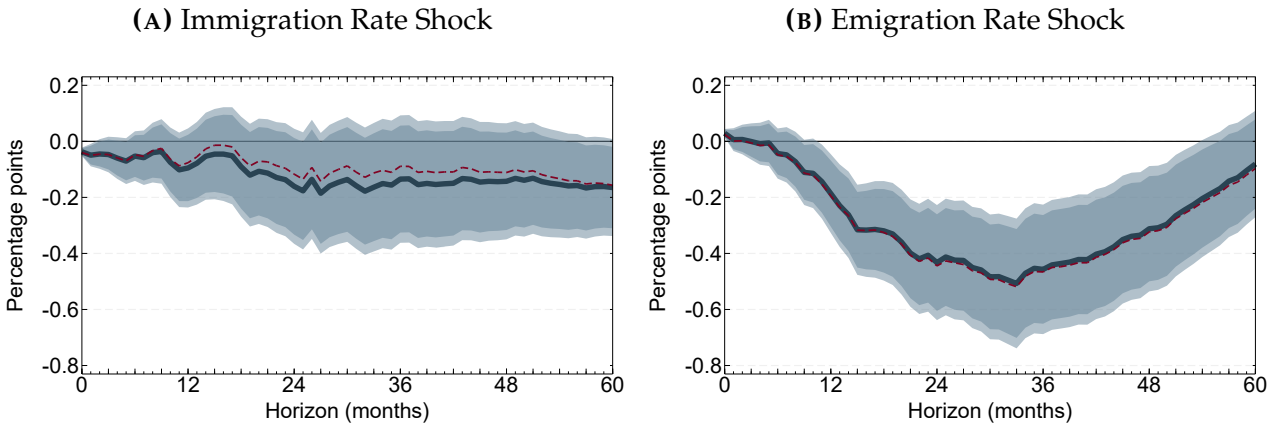
Results. I now turn to presenting and discussing the impulse response functions (IRFs) that most closely relate to the predictions of Section III. The full set of IRFs is shown in Figure A.3 in Appendix A.7. Most of the analytical discussion of Section III was on the effects and responses of emigration, and while this will also be the focus of my discussion in this subsection, I present the analogous immigration results for comparison of any (a)symmetries. All figures show IRFs for a 60-month (5-year) horizon under the variable ordering previously discussed. Shaded areas show 95% and 90% confidence bands, in which standard errors were estimated under a normal approximation of 10,000 recursive wild bootstraps. The dashed red line on the IRFs for migration shocks shows the point estimates under the alternative in which emigration is ordered before immigration in the Cholesky decomposition. All shocks

[10] In particular, while migration shift-share instruments as used by Card (2009) or Howard (2020) work well for immigration as they rely on exogenous push factors, pull factors are too weak in predicting emigration—the relevant migration flow in Section III.

are normalised to increase their corresponding variable by 1 unit on impact.

Figure 3 plots the response of the unemployment rate to a shock that raises either the immigration (panel A) or emigration (panel B) rate by 1 percentage point annualised (i.e. a monthly flow of $1/12 \approx 0.08\%$ of the population on impact). Both shocks decay quickly after impact, corresponding to a cumulative inflow of 0.40% of the population (immigration) and an outflow of 0.25% (emigration) within the first year. Swapping immigration and emigration in the Cholesky ordering has visually almost no impact on results.

FIGURE 3: Responses of Unemployment Rate (SVAR)



Notes: Shaded areas show 95% and 90% confidence bands using 10,000 recursive wild bootstraps under a normal approximation. Dashed red line plots point estimates when emigration is instead ordered before immigration in the Cholesky decomposition.

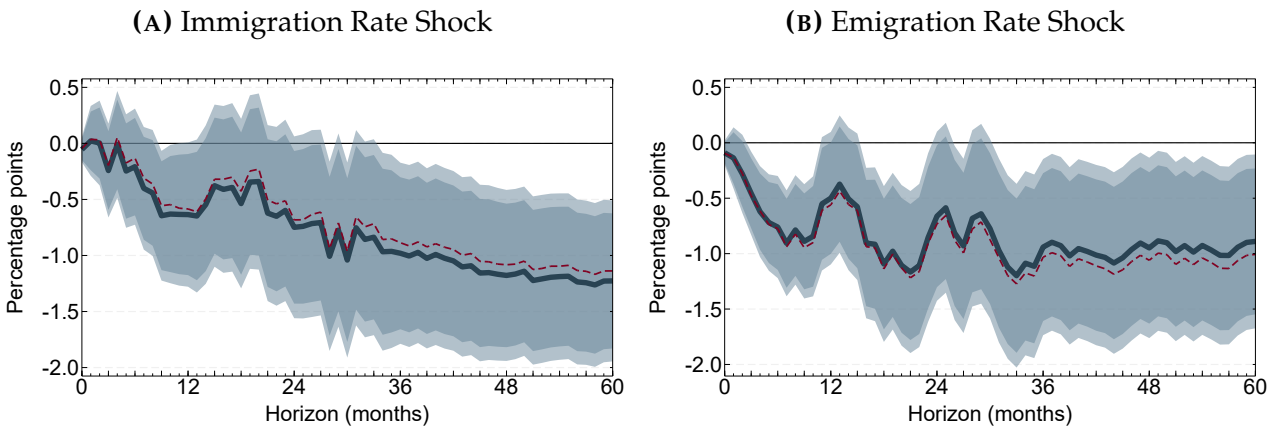
The estimates provide some evidence for the migration spillover effect from Proposition 1. A positive 1pp emigration rate shock leads to a significant reduction in unemployment in the origin region, peaking at a 0.5pp decline after almost 3 years. This can be interpreted in numerous ways. From the perspective of the model and predictions of Sections II and III, emigration raised per-capita output, which increased wages, and consequently increased labour supply. While the model did not have unemployment defined explicitly, this is consistent with the predicted fall in $1 - \ell_{i,t}$. In reality, the drop in unemployment can be an indication that the unemployed are the most likely to emigrate.

Curiously, an immigration rate shock also decreases unemployment in the receiving country, albeit with a milder and less precisely estimated 0.2pp decline. This is harder to reconcile with the symmetric setup of my model, but it could be explained by alternative directed search models, in which the unemployed in one country search for vacancies elsewhere, so successful migration reduces measured unemployment in both origin and destination.

Figure 4 plots the response of debt-to-trend-GDP to an immigration (panel A) and an emigration (panel B) shock. As before, swapping the ordering of the two migration rates in the Cholesky decomposition makes no significant difference to the point estimates.

When the annualised emigration rate exogenously increases by 1pp, debt-to-trend-GDP in the origin country falls by around 1pp after 30 months, and remains stable and persistent

FIGURE 4: Responses of Debt-to-Trend-GDP (SVAR)



Notes: Shaded areas show 95% and 90% confidence bands using 10,000 recursive wild bootstraps under a normal approximation. Dashed red line plots point estimates when emigration is instead ordered before immigration in the Cholesky decomposition.

at that level for at least 60 months (5 years). An immigration rate shock produces an effect of the same sign and similar magnitude.

These responses relate to the predictions of Proposition 3, though the mapping is not immediate. Equation (36) characterises how the origin region's fiscal space responds to emigration, with the fiscal contribution channel and the debt burden channel pulling in opposite directions. The immigration shock IRF in panel A instead captures the destination's response, in which both channels flip sign: per-capita output falls with population inflows (by Proposition 1), shrinking the per-capita tax base, while the existing debt stock is spread over more workers, easing the per-capita burden. The fact that the two IRFs are not mirror images is therefore consistent with Proposition 3 once asymmetries in labour supply parameters, spillover effects, and legacy debt servicing costs across origin and destination are accounted for. Interestingly, empirically it seems to be the case that migration reduces public debt, both for the origin and the destination country. This is consistent with the reduction in unemployment under both shocks (Figure 3), which would also indicate a positive gain in net fiscal contributions.

Most importantly, (negative) debt-to-trend-GDP is not the same object as fiscal space $\mathcal{F}_{i,t}$ as defined in Section III. Both channels of (36) can rationalise the observed fall in debt. If the fiscal contribution channel dominates, emigration raises per-capita tax revenue, primary surpluses improve, and total debt declines. If instead the debt burden channel dominates, the fiscal constraint (35) binds and forces the government into a fiscal consolidation, which also reduces total debt. Therefore, while these results are consistent with the predictions of Proposition 3, the sign of the debt response is uninformative about which channel dominates, whether labour mobility relaxes or tightens fiscal constraints, and consequently about whether fiscal-mobility interactions stabilise or amplify the original shock. Answering this question requires looking at the response of other fiscal variables, a task that I turn to next.

V.B Response of fiscal variables

Whether fiscal-mobility interactions are stabilising or amplifying depends on the sign of how migration affects fiscal space, as shown in (36). When the constraint binds and fiscal space reacts positively to emigration, the government can pursue a more counter-cyclical fiscal policy (or less pro-cyclical than otherwise) and fiscal-mobility has a stabilising role. When fiscal space instead contracts following emigration, the government is forced into deeper pro-cyclical fiscal consolidations and fiscal-mobility has an amplifying role.

Testing this in the data is complicated by the fact that the fiscal space measure $\mathcal{F}_{i,t}$ used in Section III is a model object: it represents the maximum the government can spend per capita, and depends on model parameters that cannot be directly observed. The empirical literature has instead used the (negative) debt-to-GDP ratio as a measure of fiscal space (Romer and Romer, 2019; Bellifemine et al., 2025), but the two are not the same object and, as argued in the previous subsection, the debt response is uninformative about which channel of (36) dominates.

Nonetheless, the model still provides a test of fiscal-mobility stabilisation versus amplification. According to Section III.B, a fiscally constrained government's consumption will track its fiscal space, $g_{i,t} = \mathcal{F}_{i,t}$, and the constraint is more likely to bind when the existing debt stock $b_{i,t-1}$ is high. Then, a positive response of government consumption to an emigration shock will be a sign of fiscal-mobility stabilisation, whereas a negative response signals amplification.

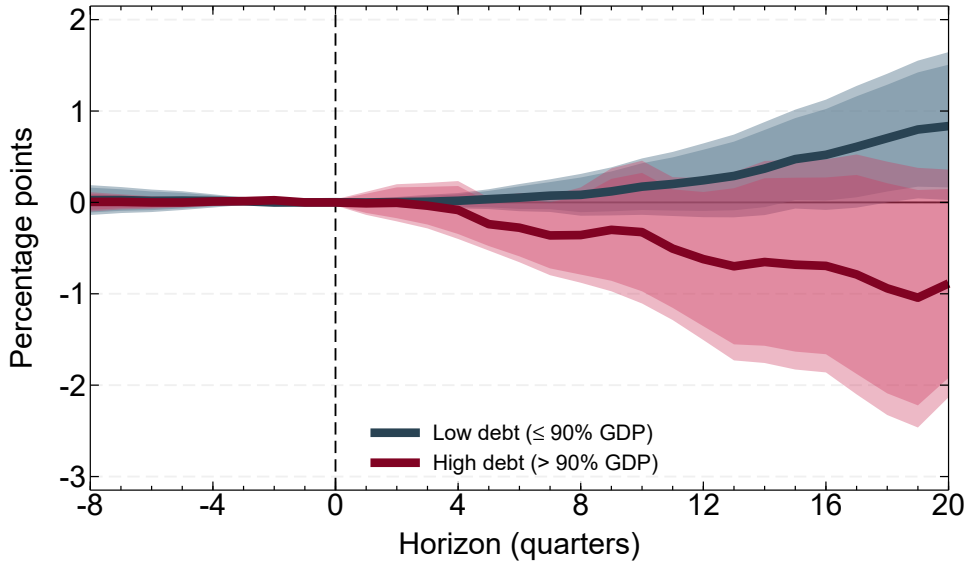
This test requires moving to quarterly frequency, since data on government consumption is not available at monthly frequency for EU countries. I obtain quarterly emigration shocks by summing the monthly emigration rate structural residuals $\hat{\varepsilon}^{Emi}_{i,t}$ from the SVAR (44) within each quarter. Using the estimated shocks, I estimate the response of government consumption $G_{i,t}$ to an emigration shock normalised to 1 standard deviation, differentially across low- and high-debt regimes, with the following local projection (Jordà, 2005),

$$\begin{aligned}\sum_{s=0}^{h} \frac{G_{i,t+s} - G_{i,t-1}}{Y_{i,t-1}} = \beta^h_L \cdot \hat{\varepsilon}^{Emi}_{i,t} \times \mathbb{1}\left\{\tfrac{B_{i,t-1}}{Y_{i,t-1}} \leq 90\%\right\} + \beta^h_H \cdot \hat{\varepsilon}^{Emi}_{i,t} \times \mathbb{1}\left\{\tfrac{B_{i,t-1}}{Y_{i,t-1}} > 90\%\right\} \\ + \mathbf{\Gamma}^{h\prime}\mathbf{X}_{i,t} + \alpha^h_i + \delta^h_{m(i,t),t} + v^h_{i,t}\end{aligned}, \tag{45}$$

where t now corresponds to a quarter instead of a month. The left-hand side denotes the cumulative change in government consumption relative to the quarter prior to the shock, scaled by pre-shock trend GDP $Y_{i,t-1}$. While the model defines fiscal space $\mathcal{F}_{i,t}$ in terms of per-capita government spending $g_{i,t}$ rather than aggregate spending $G_{i,t}$, the ratio $G_{i,t}/Y_{i,t}$ coincides with the per-capita ratio $g_{i,t}/y_{i,t}$, the model-relevant object. The coefficients of interest are β^h_L and β^h_H, capturing the cumulative response of government consumption at each horizon h in low- and high-debt regimes, respectively. I define the high-debt regime as debt $B_{i,t-1}$ exceeding 90% of trend GDP, a threshold taken from Reinhart and Rogoff (2010). Results are robust and more pronounced when instead using the higher 120% limit considered

by Herndon et al. (2014) (see Figure A.4 in Appendix A.8). Although the emigration shocks are by construction orthogonal to all lags of the endogenous variables in the SVAR (44), I include in $\mathbf{X}_{i,t}$ a vector of controls comprising the direct effect of the high-debt dummy and twelve quarterly lags of the debt-to-trend-GDP ratio (matching the three years of lags in the SVAR), to absorb residual variation in debt dynamics that might otherwise be conflated with the regime split. Results are robust to removing these lags, although confidence bands widen. As in the SVAR (44), α_i^h is a country fixed effect, and $\delta^h_{m(i,t),t}$ is a time fixed effect that varies with EU and Euro Area membership status. Standard errors are clustered at the country level.

FIGURE 5: Cumulative Response of Government Consumption to an Emigration Rate Shock (LP)



Notes: Figure shows the cumulative response of government consumption, expressed as a percentage of pre-shock trend GDP. The shock is the (unscaled) emigration rate structural residual from (44), therefore corresponding to a positive 1 standard deviation shock. Shaded areas show 95% and 90% confidence bands using standard errors clustered at the country level.

Figure 5 plots the impulse response functions under the two debt regimes. The absence of pre-trends in spending, which is not included in the SVAR and so is not mechanically orthogonal to the shocks, suggests the identified shocks are not predictable from past spending, mitigating concerns about fiscal foresight or anticipation effects. In low-debt regimes, an emigration shock raises cumulative government consumption by 0.8pp of trend GDP over 20 quarters. This sign is at odds with the prediction of an emigration shock to an unconstrained country (emigration raises per-capita output, and the government should counter-cyclically cut consumption), but is compatible with a government whose consumption is still affected by its fiscal space, while its low level of debt implies emigration has a net positive effect on fiscal space allowing it to increase spending—i.e. a government in the fiscal-mobility stabilisation region of Figure 2.

In high-debt regimes, the cumulative response is -0.9pp, though imprecisely estimated, consistent with emigration contracting fiscal space and forcing fiscal consolidations. What the data identifies sharply is the contrast between the two: the cumulative responses diverge

persistently after 10 quarters at the 10% level, reaching a difference of 1.7pp of trend GDP by quarter 20, significant at the 5% level. The regime differential is the object the model speaks to. Equation (36) predicts that the net effect of emigration on fiscal space falls as legacy debt rises, because the debt burden channel scales with $b_{i,t-1}$ while the fiscal contribution channel, for a given migration spillover, does not. A 1.7pp difference between regimes is direct evidence of that mechanism.

Peripheral countries' debt breached these debt thresholds during the sovereign debt crisis, making the results direct evidence of fiscal-mobility amplification plaguing highly indebted countries in Europe during this period, by further shrinking their fiscal space and deepening pro-cyclical consolidations. This indicates that the stabilising interaction between mobility and fiscal space is absent precisely in the states where fiscal stabilisation matters most, so emigration cannot be relied on to ease the constraints of a fiscally distressed member state.

VI Conclusion

The shock propagation role of labour mobility in a currency union is fundamentally tied to the fiscal space of its member governments, an interdependence that the existing literature, in treating labour mobility and fiscal policy as separable adjustment mechanisms, has so far overlooked. When workers emigrate from a depressed region, they generate spillover effects that raise per-capita fiscal contributions, but they also spread the existing debt stock over fewer taxpayers, raising the per-capita debt burden. The dominant channel determines whether fiscal-mobility interactions reinforce labour mobility stabilisation or amplify the original shock. This means that, for a given migration spillover, countries with higher debt should be more likely to be in an amplifying regime. I confirm this prediction empirically using a novel monthly panel of intra-EU migration flows for 2000–2019, showing that in high-debt regimes emigration reduces cumulative government consumption by 1.7pp of trend GDP over 20 quarters, relative to low-debt regimes.

These results have policy implications that should provide a new lens through which we view and interpret austerity programmes in currency unions, with a clear focus on the Euro Area. The conventional framing during the sovereign debt crisis is that austerity was necessary for the stability of the union, while emigration was a natural outcome which was separately helpful in union-wide stabilisation, but this interpretation misses a key feedback loop. Emigration from fiscally constrained southern European economies likely deepened the required fiscal consolidations and prolonged the associated recessions, in direct conflict with the simultaneous Mundellian adjustment effects. Austerity and emigration are not parallel adjustment mechanisms, but rather mutually reinforcing forces.

This naturally raises normative policy questions for future research. The first is quantitative: what is the welfare-optimal degree of labour mobility in a currency union, once the Mundellian stabilisation benefit is weighed against fiscal-mobility propagation? The second is institutional: can fiscal-mobility amplification be avoided under a federal fiscal

architecture, where a supranational authority distributes fiscal burdens to compensate for the loss of an emigrating workforce, thereby eliminating the potential negative impact of mobility on stabilisation? The question this paper answers—whether fiscal-mobility propagation is stabilising or amplifying—is the necessary foundation to answer such normative questions, and should frame how we think about optimal currency area design going forward.

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A Appendix

A.1 Proof of tax neutrality on migration decisions on impact

Proof. Consider a marginal change in region i's labour tax, $\mathrm{d}\tau_{i,t}$, on impact, starting from a symmetric equilibrium. This holds in particular at the initial steady state, where $a_0(s^0) = a_0$ for all agents. I further assume that the debt limit $\bar{b}_{i,t}$ does not bind, so that government spending follows the fiscal rule, $g_{i,t} = g_0 - \rho_g(y_{i,t} - y_0) + \zeta^g_{i,t}$, and that the equilibrium on impact is locally unique.

I conjecture that, following $\mathrm{d}\tau_{i,t}$, populations $n_{i,t}$, regional output $Y_{i,t}$, wages $W_{i,t}$ and government spending $g_{i,t}$ are unchanged in both regions, and verify that all equilibrium conditions continue to hold.

Step 1 (Labour market). Combining labour supply (6), labour demand (17) and labour market clearing (26) gives

$$W_{i,t} = \psi \left(\frac{y_{i,t}}{A} \right)^{\varphi}, \tag{A.1}$$

and

$$\ell_t(i) = \frac{y_{i,t}}{A}, \tag{A.2}$$

neither of which contains $\tau_{i,t}$. For given $y_{i,t}$, the economic incidence of the tax falls fully on firms.

Step 2 (Incidence). From (18) and (19), for given $y_{i,t}$,

$$\mathrm{d}\pi_t = -W_{i,t} L_{i,t} \, \mathrm{d}\tau_{i,t},$$

which is common to all residence histories s^t. Government spending depends on $\tau_{i,t}$ only through $y_{i,t}$, so $\mathrm{d}g_{i,t} = 0$ under the conjecture, and by (21) the change in revenue is absorbed by end-of-period debt,

$$\mathrm{d}b_{i,t} = -W_{i,t} \ell_t(i) \, \mathrm{d}\tau_{i,t}.$$

Step 3 (Households). Since there are no moving costs, the flow utility depends on the residence history only through the current region s_t, and the budget constraint (3) at $t+1$ depends on s^t only through $a_t(s^t)$. Hence the continuation value $\beta \mathbb{E}_t U(s^{t+1})$ depends on s^t only through $a_t(s^t)$. At a symmetric equilibrium, $W_{N,t} = W_{S,t}$, $n_{N,t} = n_{S,t}$ and $\xi_{N,t} = \xi_{S,t}$, so every agent faces an identical problem whether it resides in N or in S. Two implications follow.

(a) *Location.* Every agent is indifferent between the two regions. Under the conjecture, $W_{i,t}$ and $n_{i,t}$ are unchanged, as is the exogenous $\xi_{i,t}$, so every agent remains indifferent after the tax change, and the conjectured populations are consistent with the location choice (9).

(b) *Consumption.* The tax affects all agents through the same variables: $\mathrm{d}\pi_t$ from Step 2, the central bank's response $\mathrm{d}r_t$, and any change in future variables. Hence $\mathrm{d}c_t(s^t)$ depends on s^t only through $a_{t-1}(s^{t-1})$. Since residents of the two regions have the same distribution

of asset holdings at a symmetric equilibrium,

$$\sum_{s^t|s_t=N} n(s^t)\,\mathrm{d}c_t(s^t) = \sum_{s^t|s_t=S} n(s^t)\,\mathrm{d}c_t(s^t).$$

Step 4 (Goods market). At a symmetric equilibrium, $\partial Y_t/\partial Y_{i,t}$ in (24) is equal across regions, so the central bank's policy of keeping Y_t fixed implies $\mathrm{d}Y_{N,t} + \mathrm{d}Y_{S,t} = 0$ to first order. Summing the goods market clearing condition (25) across regions, with $g_{N,t}$ and $g_{S,t}$ unchanged, union-wide private consumption is therefore unchanged. Combined with Step 3(b),

$$\sum_{s^t|s_t=i} n(s^t)\,\mathrm{d}c_t(s^t) = 0, \qquad i \in \{N, S\}.$$

With fixed prices in (8), each resident's demand for each region's good is a fixed share of its consumption, so (25) gives $\mathrm{d}Y_{N,t} = \mathrm{d}Y_{S,t} = 0$. Per-capita output $y_{i,t}$ is then unchanged in both regions, and so, by Step 1, are wages $W_{i,t}$ and, by the fiscal rule, government spending $g_{i,t}$. This verifies the conjecture.

The conjectured allocation therefore remains an equilibrium following the tax change. By local uniqueness, it is the equilibrium, so

$$\frac{\mathrm{d}n_{i,t}}{\mathrm{d}\tau_{i,t}} = 0,$$

which proves (28). □

A.2 Proof of Proposition 1

Proof. As in the rest of Section III, prices are fixed at one on impact, $\tau_{i,t} = \tau_0$, and the central bank keeps union-wide output Y_t in (24) fixed. Define

$$C_{i,t} \equiv \sum_{s^t|s_t=i} n(s^t)c_t(s^t), \qquad c_{i,t} \equiv \frac{C_{i,t}}{n_{i,t}},$$

as aggregate and average consumption of region i's residents. All derivatives are evaluated at a symmetric baseline: $n_{N,t} = n_{S,t} = \frac{1}{2}$, $\alpha_{N,t} = \alpha_{S,t}$, $g_{N,t} = g_{S,t}$, and the same distribution of assets $a_{t-1}(s^{t-1})$ across residents in both regions. The baseline need not be a steady state, and the levels of assets and debt are unrestricted. In particular, public debt need not be symmetric across regions: $b_{N,t-1}$ and $b_{S,t-1}$ may differ, as long as both governments set the same per-capita spending at the baseline. I assume that additional residents are representative, in that their consumption at the baseline equals the regional average.

Step 1 (Consumption response). With $P_{i,t} = 1$, labour demand (17) and labour market clearing (26) give $\ell_{i,t} = y_{i,t}/A$, and labour supply (6) then gives the wage (32), $W_{i,t} = \psi\,(y_{i,t}/A)^{\varphi}$.

Hence $W_{i,t}\ell_{i,t} = \psi\,(y_{i,t}/A)^{1+\varphi}$ and, using $W_{i,t}\ell_{i,t}/y_{i,t} = W_{i,t}L_{i,t}/Y_{i,t}$,

$$\frac{\mathrm{d}(W_{i,t}\ell_{i,t})}{\mathrm{d}y_{i,t}} = (1+\varphi)\frac{W_{i,t}L_{i,t}}{Y_{i,t}}. \tag{A.3}$$

At the symmetric baseline, Y_t in (24) responds equally to output in the two regions, and its population terms cancel because $\mathrm{d}n_{-i,t} = -\mathrm{d}n_{i,t}$. Keeping Y_t fixed therefore implies $\mathrm{d}Y_{N,t} + \mathrm{d}Y_{S,t} = 0$ to first order. Since $Y_{N,t} + Y_{S,t} = n_{i,t}y_{i,t} + n_{-i,t}y_{-i,t}$ and $n_{N,t} + n_{S,t} = 1$,

$$\frac{\mathrm{d}y_{-i,t}}{\mathrm{d}n_{i,t}} = -\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}}. \tag{A.4}$$

By (18), profits per resident, $\Pi_{i,t}/n_{i,t} = y_{i,t} - (1+\tau_0)\psi\,(y_{i,t}/A)^{1+\varphi}$, depend only on $y_{i,t}$ and are equal across regions at the baseline. By (19) and (A.4), union-wide per-capita profits π_t therefore do not change to first order.

Using (6) to write $\psi\ell_{i,t}^{1+\varphi} = W_{i,t}\ell_{i,t}$, the budget constraint (3) of a resident of region i becomes

$$\tilde{c}_t(s^t) + a_t(s^t) = \frac{\varphi}{1+\varphi}W_{i,t}\ell_{i,t} + \pi_t + (1+r_{t-1})a_{t-1}(s^{t-1}), \qquad s_t = i. \tag{A.5}$$

Since r_{t-1} and $a_{t-1}(s^{t-1})$ are pre-determined, its right-hand side changes by $\varphi\frac{W_{i,t}L_{i,t}}{Y_{i,t}}\mathrm{d}y_{i,t}$ for every resident, whatever their assets. I assume that, on impact and given r_t, a resident consumes a fraction between zero and one of this change, as under consumption smoothing through the Euler equation (7). Since $c_t(s^t) = \tilde{c}_t(s^t) + W_{i,t}\ell_{i,t}/(1+\varphi)$, each resident's consumption response, and hence the average response $\partial c_{i,t}/\partial y_{i,t}$ at fixed $n_{i,t}$, satisfies

$$\frac{W_{i,t}L_{i,t}}{Y_{i,t}} \le \frac{\partial c_{i,t}}{\partial y_{i,t}} \le (1+\varphi)\frac{W_{i,t}L_{i,t}}{Y_{i,t}}. \tag{A.6}$$

Since asset distributions coincide across regions, the average response is the same in region $-i$.

Step 2 (Goods market). With all prices equal to one, demands (8) and goods market clearing (25) give, in per-capita terms,

$$n_{i,t}y_{i,t} = \alpha_{i,t}n_{i,t}c_{i,t} + (1-\alpha_{-i,t})n_{-i,t}c_{-i,t} + n_{i,t}g_{i,t}. \tag{A.7}$$

At the baseline, (A.7) gives $c_{i,t} = c_{-i,t} = y_{i,t} - g_{i,t}$. Differentiate (A.7) with respect to $n_{i,t}$, holding $g_{i,t}$ and r_t fixed. An additional resident adds $c_{i,t}$ to consumption in region i and removes it from region $-i$, while incumbents' average consumption responds through $y_{i,t}$ and, by (A.4), through $y_{-i,t}$. Hence

$$y_{i,t} + n_{i,t}\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}} = (\alpha_{N,t} + \alpha_{S,t} - 1)\left[c_{i,t} + n_{i,t}\frac{\partial c_{i,t}}{\partial y_{i,t}}\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}}\right] + g_{i,t}. \tag{A.8}$$

The consumption changes in the two regions offset, so union-wide demand is unchanged and the central bank has no reason to move r_t, confirming $\mathrm{d}r_t/\mathrm{d}n_{i,t} = 0$. Substituting $c_{i,t} = y_{i,t} - g_{i,t}$ into (A.8) and solving,

$$\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}} = -\frac{(2-\alpha_{N,t}-\alpha_{S,t})(y_{i,t}-g_{i,t})}{n_{i,t}\left[1-(\alpha_{N,t}+\alpha_{S,t}-1)\dfrac{\partial c_{i,t}}{\partial y_{i,t}}\right]}. \tag{A.9}$$

Step 3 (Signing). The numerator is positive, since $\alpha_{i,t} < 1$ and $y_{i,t} - g_{i,t} = c_{i,t} > 0$. Therefore

$$\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}} < 0 \quad \Longleftrightarrow \quad (\alpha_{N,t}+\alpha_{S,t}-1)\frac{\partial c_{i,t}}{\partial y_{i,t}} < 1. \tag{A.10}$$

The left-hand side of the condition is the slope of demand for region i's goods, the right-hand side of (A.7), with respect to $Y_{i,t}$ at fixed populations. The condition is therefore the regional analogue of Keynesian-cross stability: demand for region i's goods must rise less than one-for-one with its output.

Under Keynesian-cross stability. Following the correspondence principle of Samuelson (1941, 1947), restrict attention to equilibria that are stable under a quantity-adjustment process in which, at fixed populations, region i's output rises whenever demand for its goods exceeds it. The condition in (A.10) then holds by assumption, so

$$\frac{\mathrm{d}y_{i,t}}{\mathrm{d}n_{i,t}} < 0. \tag{A.11}$$

Allowing for Keynesian-cross instability. If stability is not imposed, sufficient parametric restrictions ensure (A.10) by ruling out the unstable case. If $\alpha_{N,t} + \alpha_{S,t} \leq 1$, the left-hand side of the condition is non-positive. Otherwise, (A.6) and non-negative profits, $(1+\tau_0)W_{i,t}L_{i,t} \leq Y_{i,t}$, as imposed in Section II, give

$$(\alpha_{N,t}+\alpha_{S,t}-1)\frac{\partial c_{i,t}}{\partial y_{i,t}} \leq (\alpha_{N,t}+\alpha_{S,t}-1)\frac{1+\varphi}{1+\tau_0},$$

so (A.11) holds whenever average home bias satisfies

$$\frac{\alpha_{N,t}+\alpha_{S,t}}{2} < \alpha^* \equiv \frac{1}{2} + \frac{1+\tau_0}{2(1+\varphi)}.$$

Since $\alpha^* > \frac{1}{2}$, this allows for home bias, and if $\tau_0 \geq \varphi$ the condition holds for any degree of home bias. In either case, an increase in population in period t strictly decreases per-capita output at any symmetric baseline, completing the proof. □

A.3 Sign and magnitude of the partial-equilibrium fiscal multiplier

Consider an increase in region i's per-capita government spending, $\mathrm{d}g_{i,t} > 0$, holding populations and $g_{-i,t}$ fixed. As in Appendix A.2, I evaluate the response at a symmetric

baseline: $n_{N,t} = n_{S,t} = \frac{1}{2}$, $\alpha_{N,t} = \alpha_{S,t}$, $g_{N,t} = g_{S,t}$, and the same distribution of assets across residents in both regions. Home bias is unrestricted. Using the notation of Appendix A.2, the goods market clearing condition is

$$Y_{i,t} = \alpha_{i,t} C_{i,t} + (1 - \alpha_{-i,t}) C_{-i,t} + n_{i,t} g_{i,t}. \tag{A.12}$$

Crowding out. The central bank keeps Y_t in (24) fixed, which at the symmetric baseline implies $\mathrm{d}Y_{N,t} + \mathrm{d}Y_{S,t} = 0$ to first order. With fixed populations, $\mathrm{d}y_{-i,t} = -\mathrm{d}y_{i,t}$, so union-wide per-capita profits π_t do not change to first order, as in Appendix A.2. Summing (A.12) across regions, private consumption must be crowded out one-for-one with the increase in aggregate government spending,

$$\mathrm{d}C_{N,t} + \mathrm{d}C_{S,t} = -n_{i,t}\, \mathrm{d}g_{i,t}. \tag{A.13}$$

The central bank achieves this by raising r_t, which lowers consumption through the Euler equation (7). As in Step 3 of Appendix A.1, this effect, and that of any change in union-wide future variables, depends on a household's residence history only through its assets. Since the distribution of assets is the same in both regions, it lowers average consumption $c_{i,t} = C_{i,t}/n_{i,t}$ by the same amount in both regions.

Income response. In addition, the change in regional output changes residents' current wage surplus. As in Appendix A.2, average consumption in each region responds to its own per-capita output by $\partial c_{i,t}/\partial y_{i,t}$, which satisfies (A.6) and is the same in both regions. Combining the common crowding-out term, the income response, $\mathrm{d}y_{-i,t} = -\mathrm{d}y_{i,t}$ and (A.13) with $n_{N,t} = n_{S,t} = \frac{1}{2}$,

$$\mathrm{d}c_{i,t} = \frac{\partial c_{i,t}}{\partial y_{i,t}}\, \mathrm{d}y_{i,t} - \frac{\mathrm{d}g_{i,t}}{2}, \qquad \mathrm{d}c_{-i,t} = -\frac{\partial c_{i,t}}{\partial y_{i,t}}\, \mathrm{d}y_{i,t} - \frac{\mathrm{d}g_{i,t}}{2}. \tag{A.14}$$

Residents of region i therefore cut their consumption by less than residents of region $-i$, and may even raise it.

Output response. Differentiating (A.12) with fixed populations, so that $\mathrm{d}Y_{i,t} = n_{i,t}\, \mathrm{d}y_{i,t}$, and substituting (A.14),

$$\mathrm{d}y_{i,t} = \alpha_{i,t}\, \mathrm{d}c_{i,t} + (1 - \alpha_{-i,t})\, \mathrm{d}c_{-i,t} + \mathrm{d}g_{i,t} = (\alpha_{N,t} + \alpha_{S,t} - 1) \frac{\partial c_{i,t}}{\partial y_{i,t}}\, \mathrm{d}y_{i,t} + \frac{\mathrm{d}g_{i,t}}{2}.$$

Since $\alpha_{N,t} = \alpha_{S,t}$, the common crowding-out term falls equally on both regions' goods, so half of the increase in spending is crowded out of demand for region i's goods, whatever the degree of home bias. Solving, the regional fiscal multipliers in per-capita and aggregate terms are, respectively,

$$\frac{\mathrm{d}y_{i,t}}{\mathrm{d}g_{i,t}} = \frac{1/2}{1 - (\alpha_{N,t} + \alpha_{S,t} - 1) \dfrac{\partial c_{i,t}}{\partial y_{i,t}}}, \tag{A.15}$$

$$\frac{\mathrm{d}Y_{i,t}}{\mathrm{d}g_{i,t}} = n_{i,t}\frac{\mathrm{d}y_{i,t}}{\mathrm{d}g_{i,t}} = \frac{1/4}{1-(\alpha_{N,t}+\alpha_{S,t}-1)\dfrac{\partial c_{i,t}}{\partial y_{i,t}}}. \tag{A.16}$$

Sign. The denominator is the same as in (A.9). It is positive under Keynesian-cross stability, condition (A.10), or under the parametric restriction on home bias in Appendix A.2. Under either,

$$\frac{\mathrm{d}Y_{i,t}}{\mathrm{d}g_{i,t}} > 0 \qquad \text{and} \qquad \frac{\mathrm{d}y_{i,t}}{\mathrm{d}g_{i,t}} > 0.$$

With no home bias, $\alpha_{N,t} = \alpha_{S,t} = \frac{1}{2}$, the income response drops out and the multipliers are $\frac{\mathrm{d}Y_{i,t}}{\mathrm{d}g_{i,t}} = \frac{1}{4}$ and $\frac{\mathrm{d}y_{i,t}}{\mathrm{d}g_{i,t}} = \frac{1}{2}$, which gives the coefficient on $\hat{g}_{i,t}$ in (RD). With home bias, $\alpha_{N,t} + \alpha_{S,t} > 1$, and since $\partial c_{i,t}/\partial y_{i,t} > 0$ by (A.6), both multipliers are larger: part of the additional income of region i's residents is spent on region i's own goods.

A.4 Solution to system of $(\hat{y}_{i,t}, \hat{n}_{i,t}, \hat{g}_{i,t})$

Unconstrained government.

$$\begin{bmatrix} \hat{y}_{i,t} \\ \hat{n}_{i,t} \\ \hat{g}_{i,t} \end{bmatrix} = \mathcal{M}_t^{FR} \begin{bmatrix} 1 & -\dfrac{c_t}{y_t} & \dfrac{1}{2}\dfrac{g_t}{y_t} \\ \dfrac{\kappa_t}{1-\kappa_t} & 1+\dfrac{1}{2}\rho_g & \dfrac{1}{2}\dfrac{g_t}{y_t}\dfrac{\kappa_t}{1-\kappa_t} \\ -\rho_g\dfrac{y_t}{g_t} & \rho_g\dfrac{c_t}{g_t} & 1+\dfrac{\kappa_t}{1-\kappa_t}\dfrac{c_t}{y_t} \end{bmatrix} \begin{bmatrix} \varepsilon_{i,t}^{RD} \\ \varepsilon_{i,t}^{LM} \\ \varepsilon_{i,t}^{FR} \end{bmatrix}, \tag{A.17}$$

$$\mathcal{M}_t^{FR} = \left[1 + \frac{1}{2}\rho_g + \frac{\kappa_t}{1-\kappa_t}\frac{c_t}{y_t}\right]^{-1}.$$

Constrained government with no labour mobility.

$$\begin{bmatrix} \hat{y}_{i,t} \\ \hat{n}_{i,t} \\ \hat{g}_{i,t} \end{bmatrix}_{\hat{n}_{i,t}=0} = \mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0} \begin{bmatrix} 1 & 0 & \dfrac{1}{2}\dfrac{g_t}{y_t} \\ 0 & 0 & 0 \\ (1+\varphi)\dfrac{\tau_t W_t \ell_t}{g_t} & 0 & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_{i,t}^{RD} \\ \varepsilon_{i,t}^{LM} \\ \varepsilon_{i,t}^{FS} \end{bmatrix}, \tag{A.18}$$

$$\mathcal{M}_t^{FS}\Big|_{\hat{n}_{i,t}=0} = \left[1 - \frac{1}{2}(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}\right]^{-1}.$$

Constrained government with labour mobility.

$$\begin{bmatrix}\hat{y}_{i,t}\\ \hat{n}_{i,t}\\ \hat{g}_{i,t}\end{bmatrix} = \mathcal{M}_t^{FS}\begin{bmatrix}1 & -\frac{c_t}{y_t}+\frac{1}{2}\frac{(1+r_{t-1})b_{i,t-1}}{y_t} & \frac{1}{2}\frac{g_t}{y_t}\\ \frac{\kappa_t}{1-\kappa_t} & 1-\frac{1}{2}(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t} & \frac{1}{2}\frac{g_t}{y_t}\frac{\kappa_t}{1-\kappa_t}\\ (1+\varphi)\frac{\tau_t W_t \ell_t}{g_t}+\frac{\kappa_t}{1-\kappa_t}\frac{(1+r_{t-1})b_{i,t-1}}{g_t} & \frac{(1+r_{t-1})b_{i,t-1}}{g_t}-(1+\varphi)\frac{\tau_t W_t \ell_t}{g_t}\frac{c_t}{y_t} & 1+\frac{\kappa_t}{1-\kappa_t}\frac{c_t}{y_t}\end{bmatrix}\begin{bmatrix}\varepsilon_{i,t}^{RD}\\ \varepsilon_{i,t}^{LM}\\ \varepsilon_{i,t}^{FS}\end{bmatrix},$$

$$\mathcal{M}_t^{FS} = \left[1-\frac{1}{2}(1+\varphi)\frac{\tau_t W_t \ell_t}{y_t}+\frac{\kappa_t}{1-\kappa_t}\left(\frac{c_t}{y_t}-\frac{1}{2}\frac{(1+r_{t-1})b_{i,t-1}}{y_t}\right)\right]^{-1}. \tag{A.19}$$

A.5 Details on data collection and cleaning

A.5.1 LinkedIn migration flows

I record a migration flow whenever a LinkedIn user's CV shows a new job or education spell starting in an EU country that differs from the country of their previous position, also in the EU. Flows are weighted by Revelio Labs' sampling weights (which take into account occupation and geography) to better approximate the overall population. The timing of a recorded migration move is subject to measurement error. Users who do not remember the month they started a job often leave it blank, which Revelio Labs codes as January, so the true month of migration may be later than reported. The gap can also go the other way, with workers moving a few months before their new job starts. The problem is sharper for education: although users report a start month, Revelio Labs records only the start year. Since most educational programmes begin in September, I assume all education spells start in September of the reported year. Because these timing errors recur every year, as does true seasonality (many jobs start in January or after the summer), they show up as a seasonal pattern. I therefore apply the X-13ARIMA-SEATS seasonal adjustment filter to each country's total immigration and emigration series (the sums of the raw bilateral flows). All 56 series are strictly positive and are adjusted multiplicatively. Each adjusted series is then benchmarked to the calendar-year totals of the raw series using the proportional Denton method, so annual totals are unchanged. Because LinkedIn coverage varies across countries and over time, I express migration rates not as a share of population but as an annualised share of the (weighted) number of LinkedIn users with an active job or education spell in that country and month. Since this measure fluctuates with the business cycle, I scale by its country-specific cubic *trend* rather than the raw count.

A.5.2 Sovereign debt

The highest frequency at which government debt stock data is publicly available is quarterly. Here, I describe in more detail the methods I use to infer monthly public debt. The procedure mostly involves exploiting the fact that debt securities—the largest instrument of total government debt—are widely available at monthly frequencies.

The main data source used is the ECB's and Eurostat's Government Finance Statistics (GFS) dataset. From this, I first extract quarterly debt stocks and debt-to-GDP ratios for the entire sample period of EU-27 countries (the United Kingdom is treated separately).

Next, noticing that government debt is made up of debt securities, loans, and currency and deposits, I obtain GFS data for the stock of debt securities, which constitutes on average 72% of total government debt. Crucially, the GFS provides this data at monthly frequency, albeit only starting in late 2009. In order to fill in this gap, I use the ECB's alternative CSDB-derived Securities Issues Statistics (CSEC), which allows me to obtain monthly debt securities for the remaining earlier sample period. Where monthly GFS data is available, it is preferred over CSEC, since both the monthly and quarterly GFS series are compiled under the same accounting framework, ensuring internal consistency. Despite some missing observations, GFS combined with CSEC provides 75% observational coverage of monthly debt securities for EU-28 countries between 2000 and 2019.

A caveat of the monthly debt securities data is that it is unconsolidated, meaning it includes loans from one level of government to another, as well as between different government agencies. Consolidated debt securities are available only at a quarterly frequency. To correct for this, I take the difference between the monthly unconsolidated series and the quarterly consolidated series to obtain intra-government debt at the end-of-quarter months. I linearly interpolate intra-government debt for the remaining months of the quarter and subtract this from the monthly unconsolidated debt securities series to infer monthly consolidated securities.

I follow a similar procedure for the remaining instruments. Only debt securities are available at a monthly frequency, with the remaining debt instruments—loans, and currency and deposits—again only being available at a quarterly frequency. I take the difference between total government debt and debt securities to obtain the value of other debt instruments at end-of-quarter months. I linearly interpolate this series for the remaining months in the quarter, to infer monthly values of other debt instruments. I add this to the (consolidated) monthly debt securities, to obtain monthly total government debt.

Whenever there are gaps in the monthly GFS and CSEC series, I fill them in also by linearly interpolating the quarterly values of total debt. In summary, my final monthly series of government debt uses as much directly observable monthly variation as possible—there is monthly coverage on debt securities for 75% of observations, and debt securities make up 72% of the value of total debt. The remaining gaps are filled through linear interpolation of quarterly data. GFS and CSEC do not have detailed data for the United Kingdom, and so for this country I instead use data from the World Bank's Quarterly Public Sector Debt dataset, and the whole series is linearly interpolated into monthly.

A.5.3 10-year government bond yields

The data on 10-year government bond yields is taken from Eurostat, based on the European Monetary Union convergence criterion series of long-term interest rates. The definition

of such long-term interest rates is taken as central government bond yields with residual maturity of around 10 years. The exact data series code is `[irt_lt_mcby_m]`. Regarding the methodology for countries lacking outstanding long-term sovereign debt, the metadata for this data series states:

> *A harmonised long-term interest rate for Luxembourg is presented starting mid-May 2010. Before, the Luxembourg Government did not have outstanding long-term debt securities with a residual maturity of close to ten years. Therefore, the yield on long-term bond(s) issued by a private credit institution with a residual maturity close to 10 years is presented for the period up to mid-May 2010 and is thus not fully harmonised for that period.*
>
> *For Cyprus, primary market yields are reported up to April 2015. The same applies to Bulgaria [and] Romania up to December 2005, Slovenia up to October 2003 and Lithuania up to October 2007. For Greece, no data are available for July 2015 owing to market closure.*[11]

A summary of the number of observations on bond yields for each country is shown in Table A.1. The patterns of missing data in the Eurostat series reflect institutional features

TABLE A.1: 10-year Government Bond Yield Data Coverage

Country	Obs.	Missing Period	Country	Obs.	Missing Period
Austria	240	—	Italy	240	—
Belgium	240	—	Latvia	228	2000m1–2000m12
Bulgaria	204	2000m1–2002m12	Lithuania	228	2000m1–2000m12
Croatia	169	2000m1–2005m11	Luxembourg	240	—
Cyprus	228	2000m1–2000m12	Malta	228	2000m1–2000m12
Czechia	237	2000m1–2000m3	Netherlands	240	—
Denmark	240	—	Poland	228	2000m1–2000m12
Estonia	0	full sample	Portugal	240	—
Finland	240	—	Romania	177	2000m1–2005m3
France	240	—	Slovakia	228	2000m1–2000m12
Germany	240	—	Slovenia	214	2000m1–2002m2
Greece	239	2015m7	Spain	240	—
Hungary	228	2000m1–2000m12	Sweden	240	—
Ireland	240	—	United Kingdom	240	—
			Total	**6,196**	

Notes: Sample period is January 2000 to December 2019, giving a maximum of 240 monthly observations per country. Source: Eurostat series `irt_lt_mcby_m`.

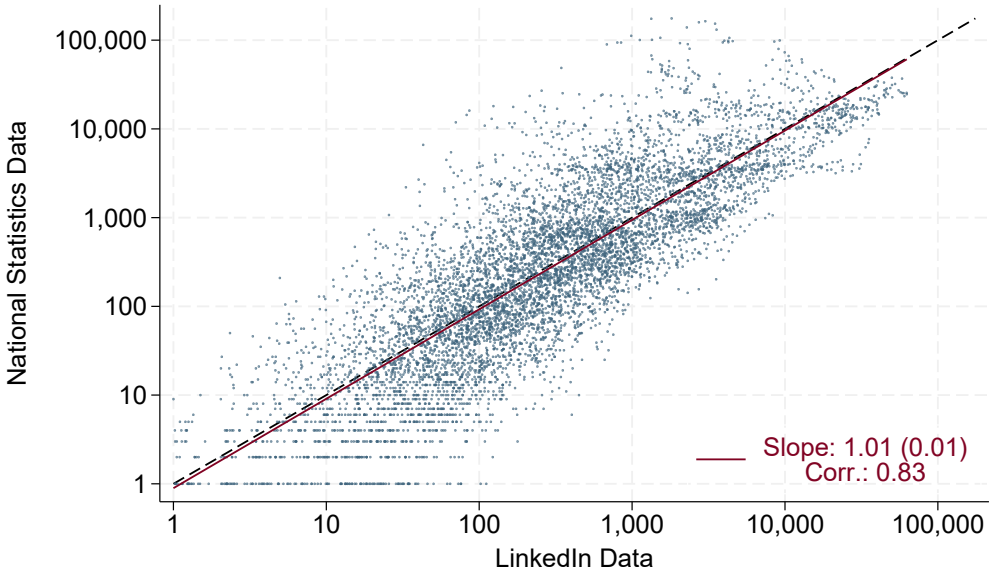
of EU accession and sovereign debt issuance. Coverage for new EU member states begins at or shortly after the start of harmonised reporting under the EMU convergence criterion framework, which typically followed the formal opening of accession negotiations. This

[11] Source: Eurostat, series `irt_lt_mcby_m`, metadata accessed 22nd May 2026. URL: https://ec.europa.eu/eurostat/cache/metadata/en/irt_lt_mcby_esms.htm.

explains the 2000 gap for Cyprus, Hungary, Latvia, Lithuania, Malta, Poland, and Slovakia (which joined the EU in 2004), as well as the longer gaps for Bulgaria and Romania (2007 accession) and Croatia (2013 accession). Estonia has no observations over the entire sample period due to its historically low public debt and limited issuance of long-term sovereign bonds—harmonised Eurostat data on Estonian 10-year yields only begins in 2021. The single missing observation for Greece (July 2015) coincides with the closure of the Athens Stock Exchange and capital controls imposed during the height of the Greek sovereign debt crisis.

A.6 Migration data visualisation

FIGURE A.1: Intra-EU28 Bilateral Migration Flows
LinkedIn vs National Statistics

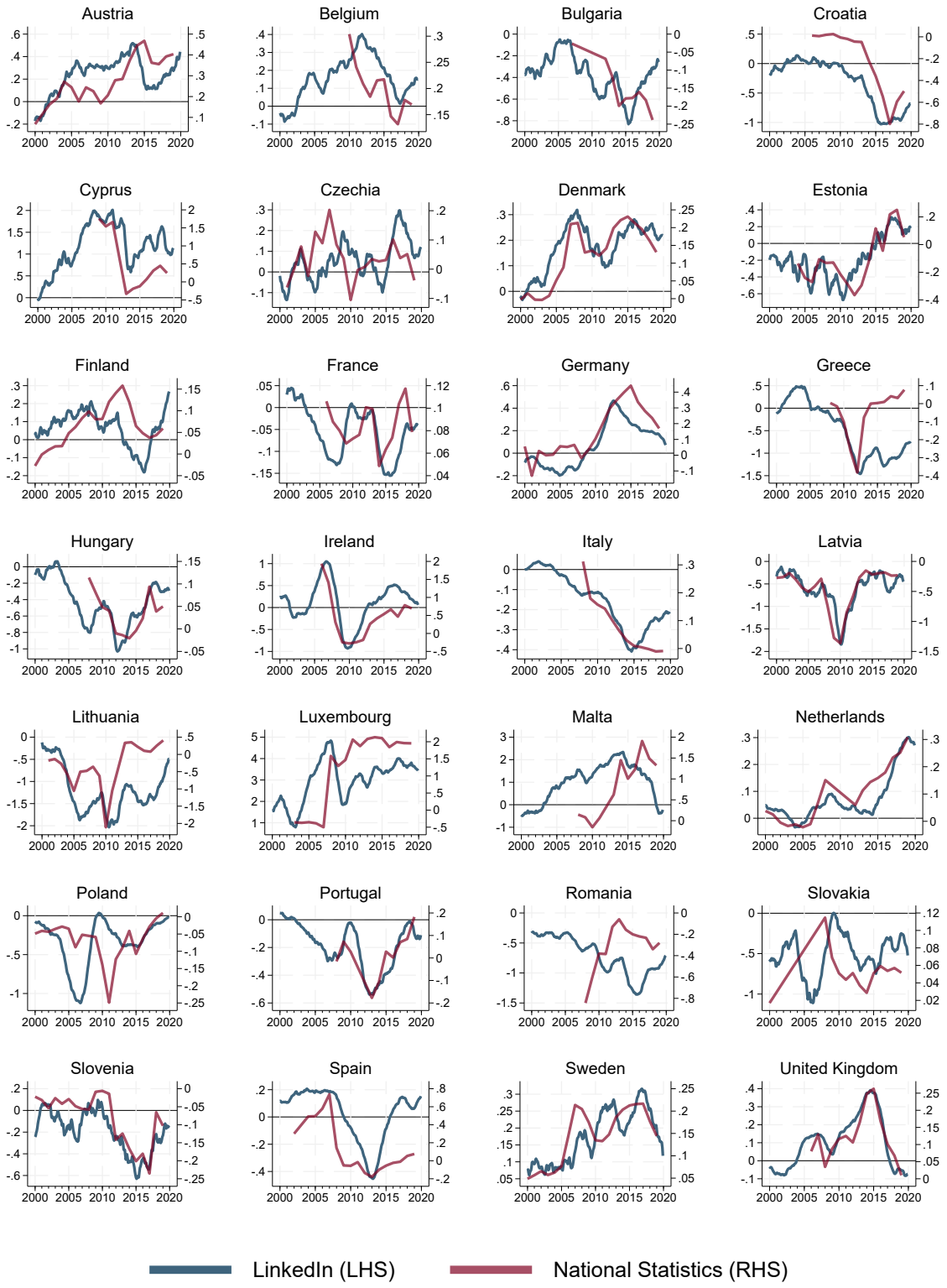


Notes: Each dot corresponds to an origin-destination-year migration flow. Dashed black line shows 45-degree line. Solid red line shows line of best fit. Regression slope coefficient (standard error in parentheses) and correlation between the two variables are displayed.

Figure A.2 plots the time series of monthly intra-EU net migration rates constructed from LinkedIn data, and the annual series from official national statistics. The two data series have overall a good match, notably for Croatia, Cyprus, Denmark, Estonia, Germany, Hungary, Ireland, Italy, Latvia, the Netherlands, Portugal, Slovenia, the UK, among others. For other countries, such as Czechia, Poland, and Romania, there is a serious mismatch.

One possible reason is simply that these national authorities do not collect data for intra-EU migration very accurately. Notice, for instance, that official statistics barely show an outflow of workers from Poland upon joining the EU in 2004, whereas the LinkedIn series shows this pattern. Countries such as Germany or the United Kingdom, which have very detailed bilateral migration data relative to other EU countries, have much closer matches. Another possibility is that the mismatched countries mostly have unskilled migration and LinkedIn is biased towards skilled workers, and the sample weights are not enough to account for the difference.

FIGURE A.2: Intra-EU28 Net Migration Rates
LinkedIn vs National Statistics

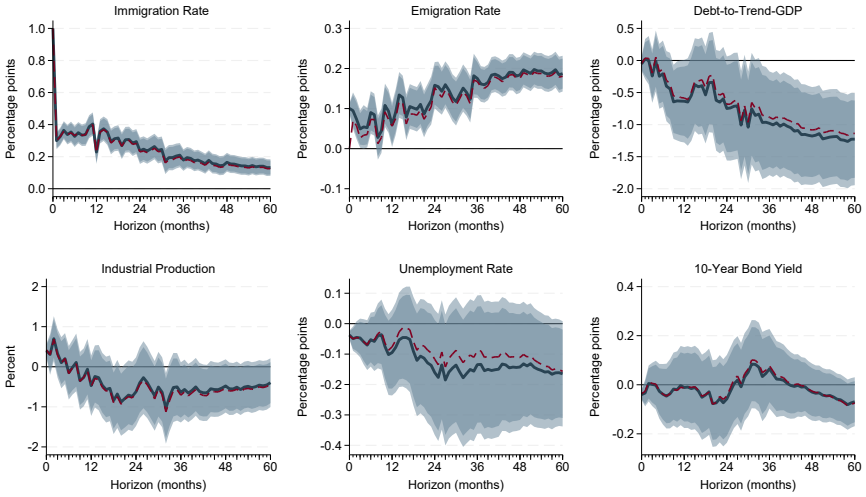


Notes: The monthly series from LinkedIn is annualised for comparability with the annual series from official national statistics, and a centred MA-12 filter was applied to it in this figure.

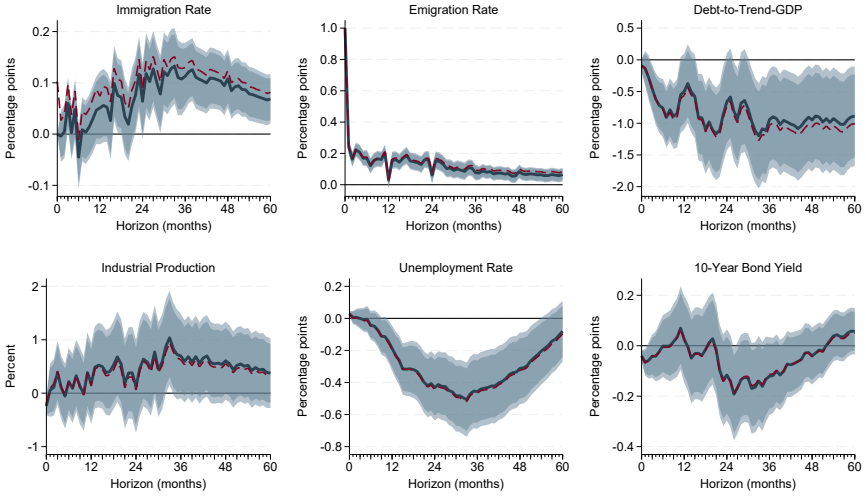
A.7 SVAR impulse response functions

FIGURE A.3: Full Set of IRFs from SVAR (44)

(A) Responses to an Immigration Rate Shock



(B) Responses to an Emigration Rate Shock



(C) Responses to a Debt-to-Trend-GDP Shock

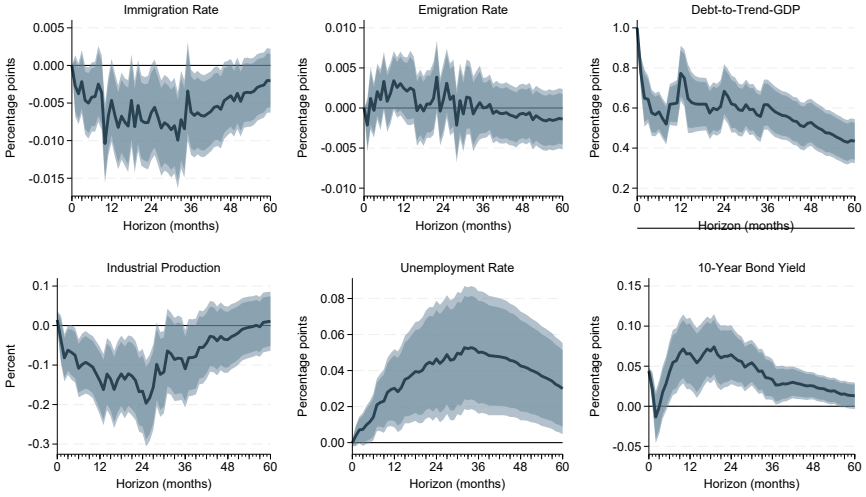
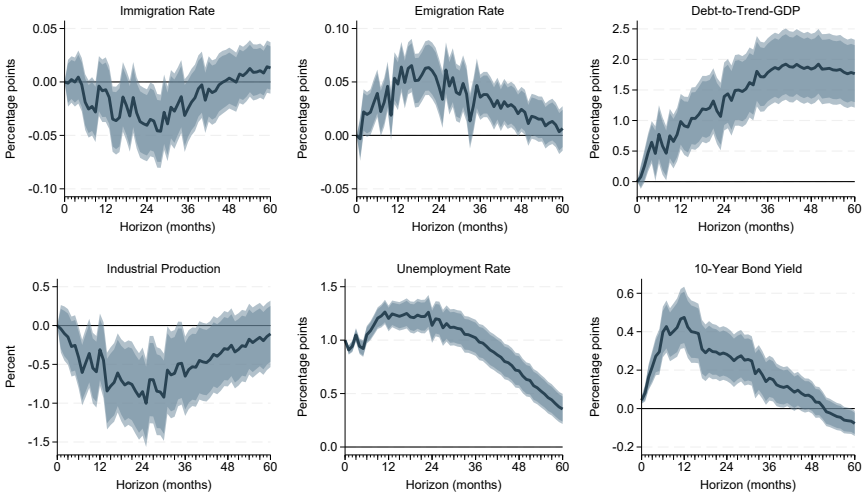
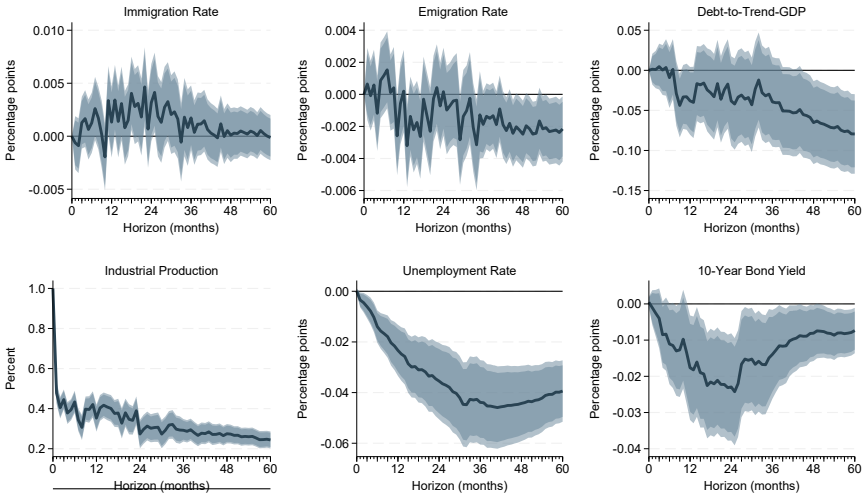


FIGURE A.3: Full Set of IRFs from SVAR (44) (continued)

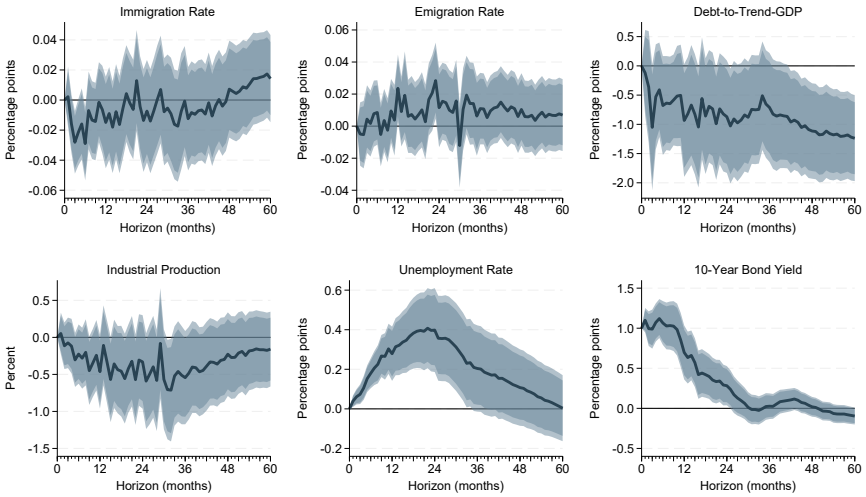
(D) Responses to an Unemployment Rate Shock



(E) Responses to an Industrial Production Shock



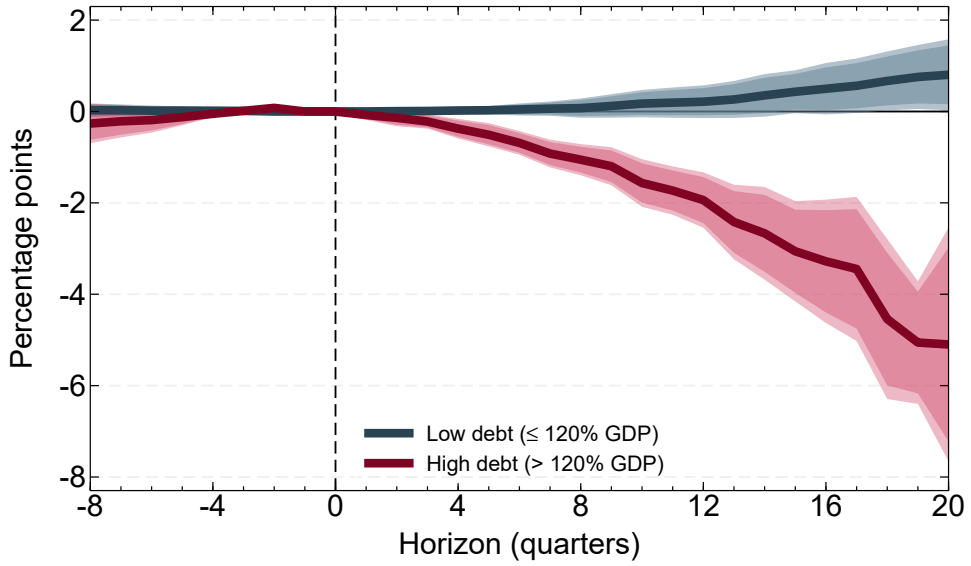
(F) Responses to a Bond Yield Shock



Notes: Shaded areas show 95% and 90% confidence bands using 10,000 recursive wild bootstraps under a normal approximation. Dashed red line plots point estimates when emigration is instead ordered before immigration in the Cholesky decomposition.

A.8 Robustness checks for fiscal variables local projection

FIGURE A.4: Cumulative Response of Government Consumption to an Emigration Rate Shock with a 120% Threshold Rule (LP)



Notes: Figure shows the cumulative response of government consumption, expressed as a percentage of pre-shock trend GDP. The shock is the (unscaled) emigration rate structural residual from (44), therefore corresponding to a positive 1 standard deviation shock. Shaded areas show 95% and 90% confidence bands using standard errors clustered at the country level.